

Can the quantum switch be deterministically simulated?

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joint work with:

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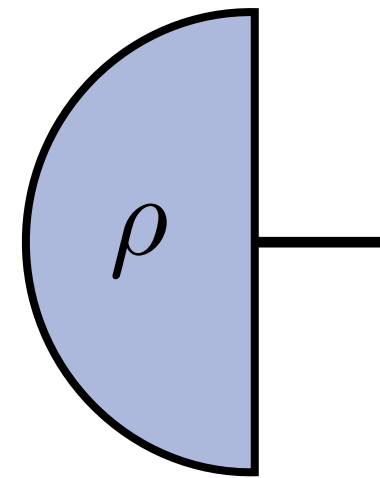
Satoshi Yoshida - University of Tokyo

Nature Communications **16**, 10216 (2025)

arXiv: 2409.18202 [quant-ph] (to be updated)

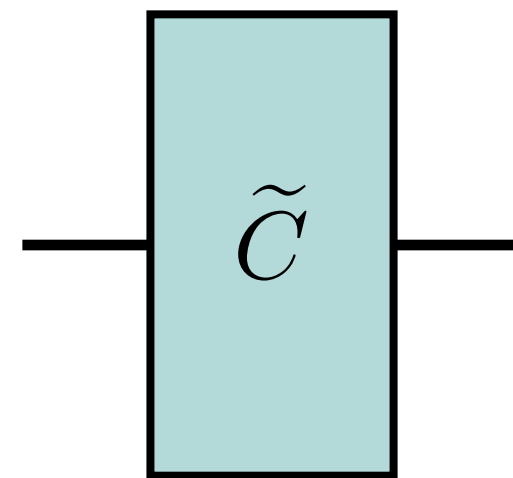
Formalism: Higher-order operations

quantum data:
quantum states

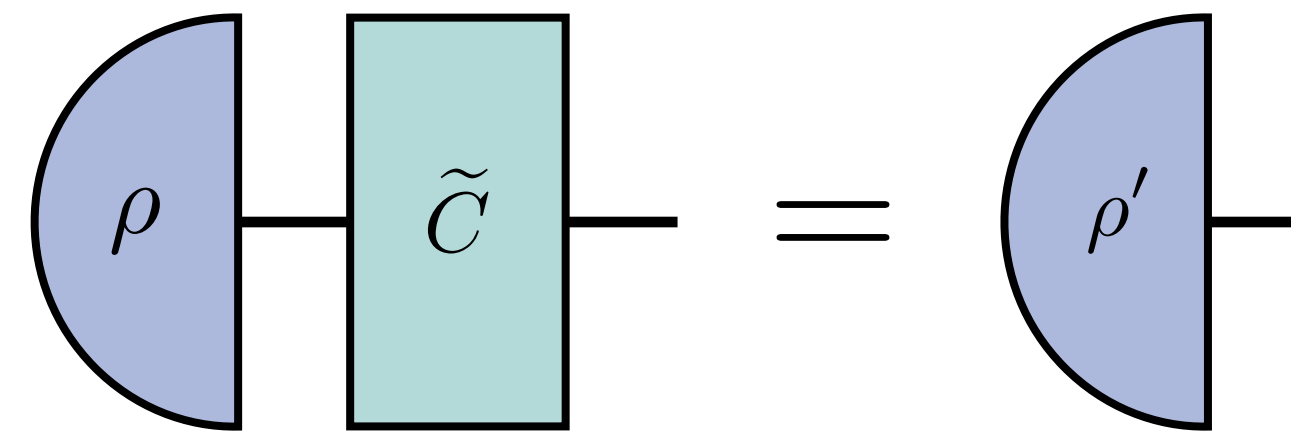


$$\rho \in \mathcal{L}(\mathcal{H}_{\text{in}})$$

quantum functions:
quantum operations
(quantum channels)

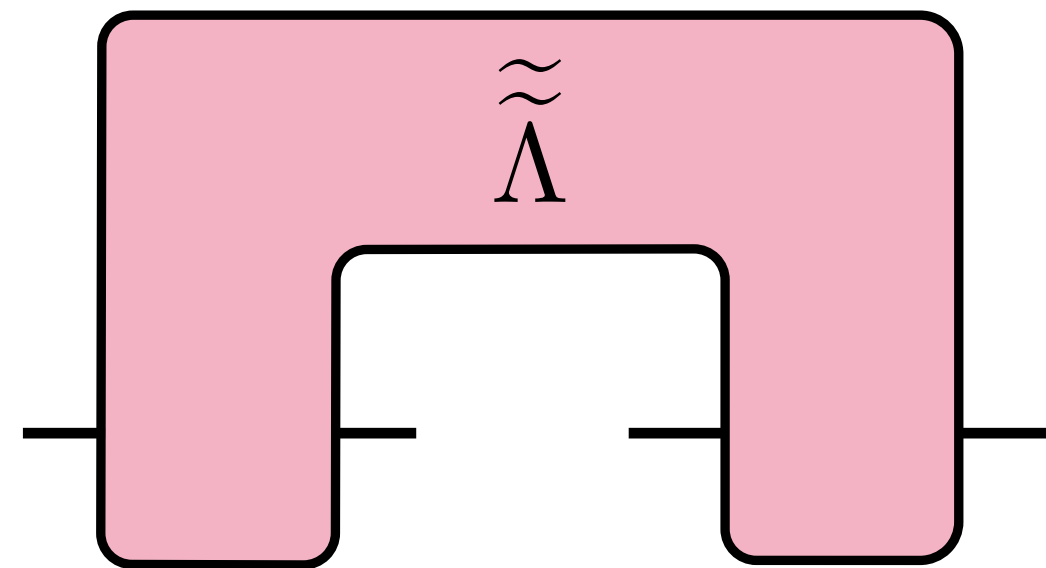


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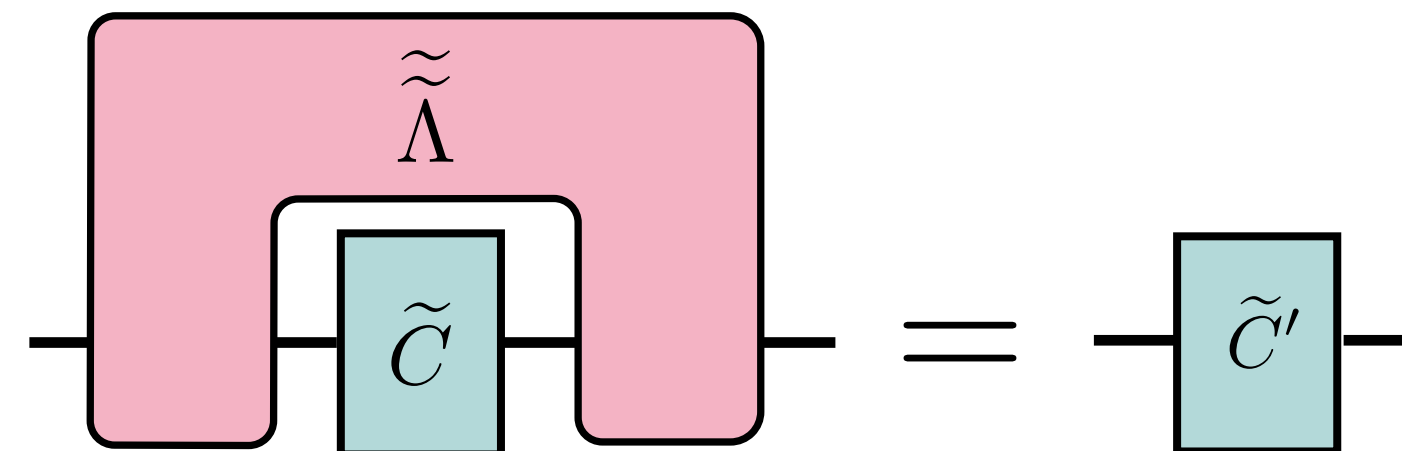


$$\tilde{C} : \mathcal{L}(\mathcal{H}_{\text{in}}) \rightarrow \mathcal{L}(\mathcal{H}_{\text{out}})$$

higher-order quantum operations:
quantum processes



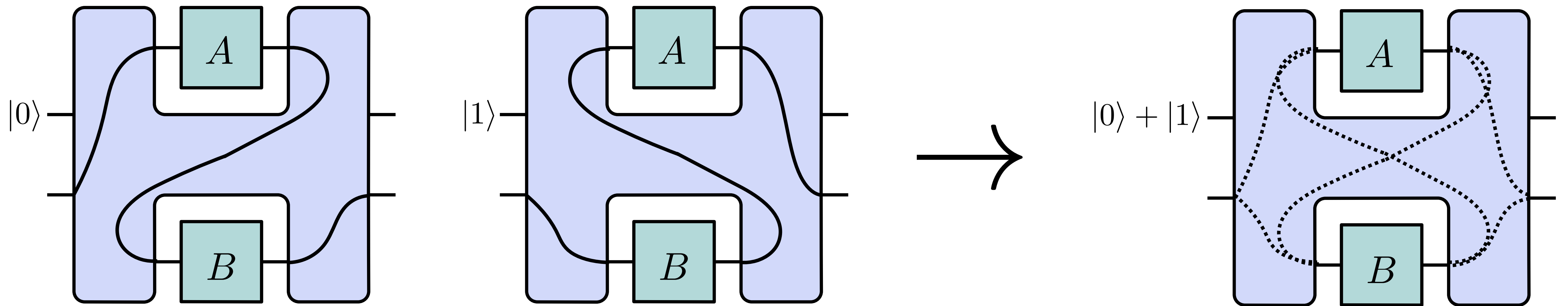
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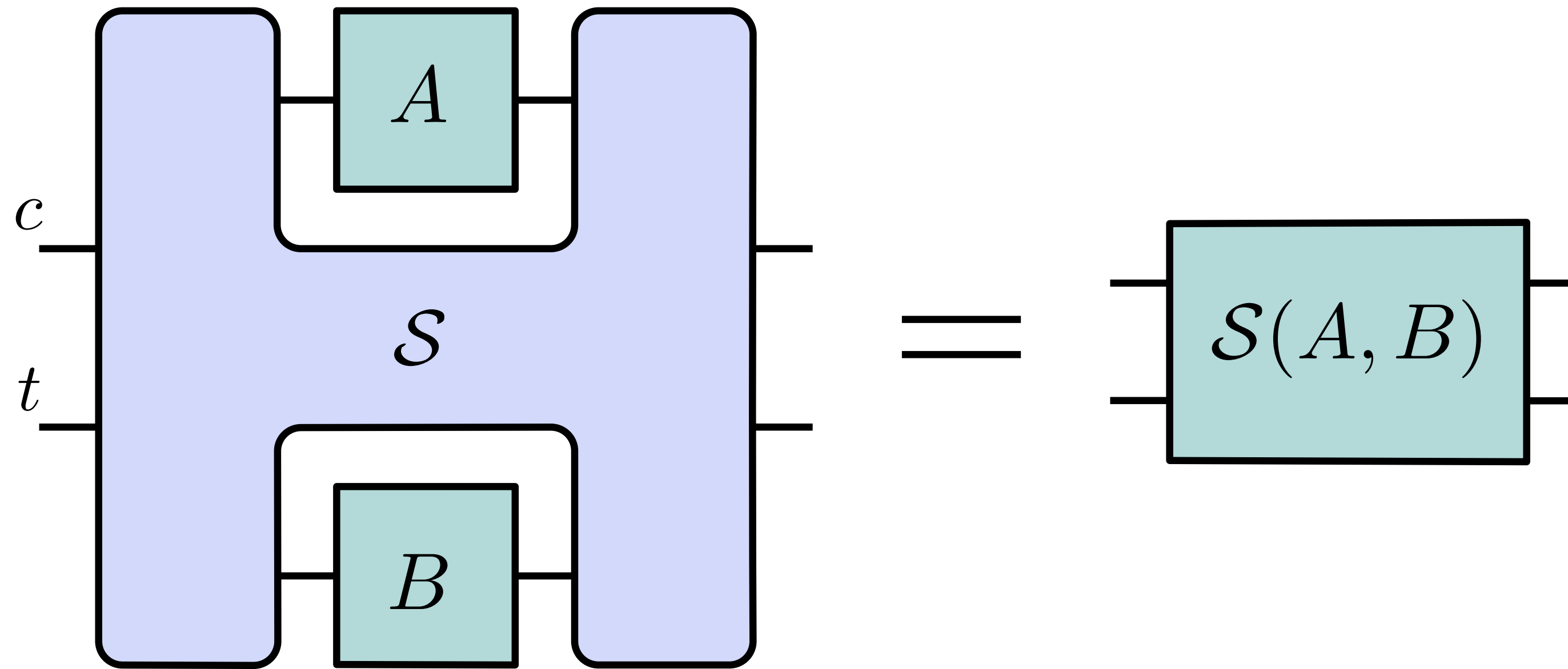
$$\begin{aligned} \tilde{\tilde{\Lambda}} : [\mathcal{L}(\mathcal{H}_{\text{in}}) \rightarrow \mathcal{L}(\mathcal{H}_{\text{out}})] \\ \rightarrow [\mathcal{L}(\mathcal{H}_{\text{in}'}) \rightarrow \mathcal{L}(\mathcal{H}_{\text{out}'})] \end{aligned}$$

“functions of functions”

The quantum switch transformation

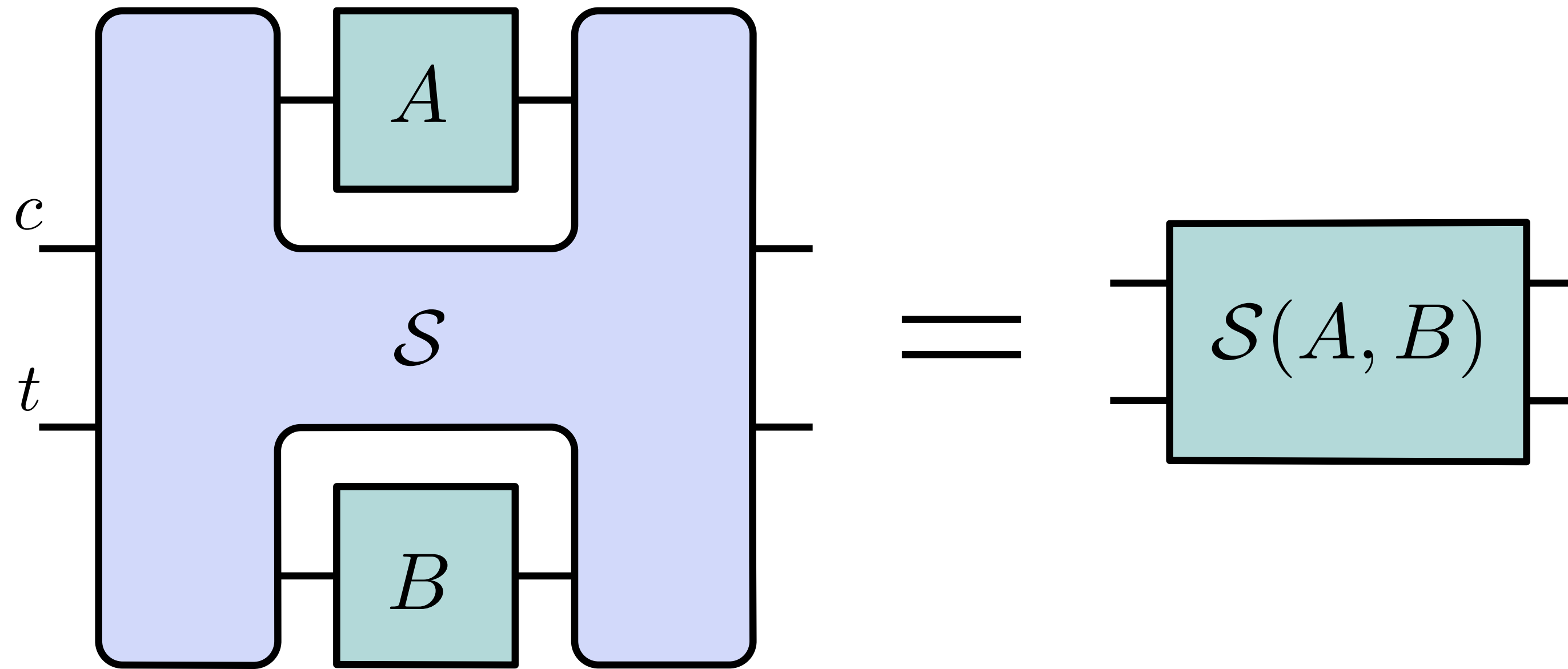


The quantum switch transformation



$$\mathcal{S}(A, B)[\sigma_c \otimes \rho_t] := \sum_{i,j} S_{ij}(\sigma_c \otimes \rho_t) S_{ij}^\dagger$$

The quantum switch transformation



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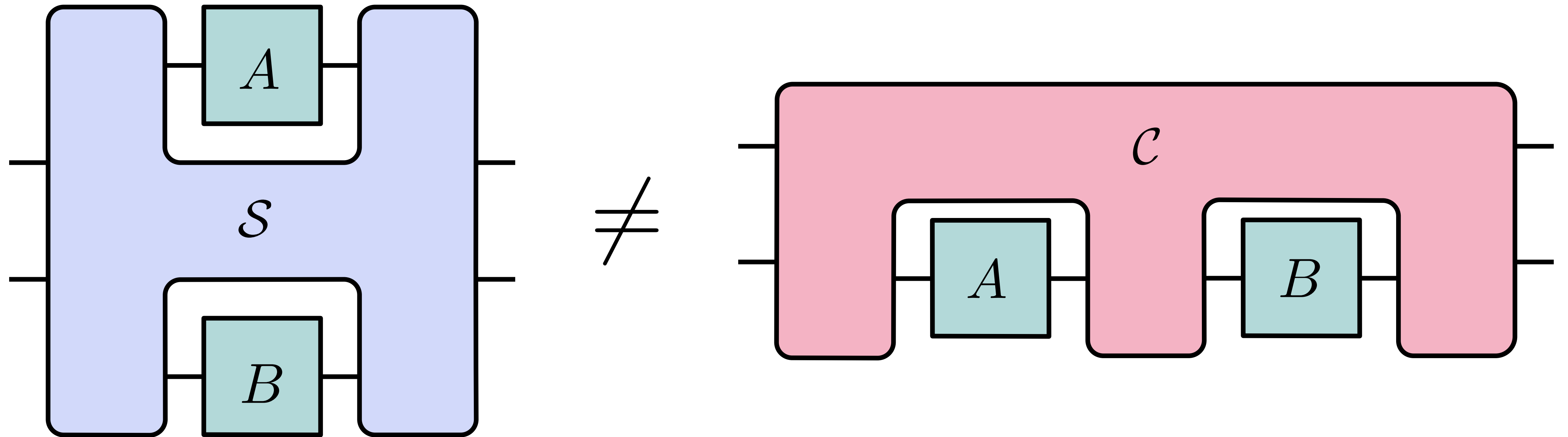
$$S_{ij} := |0\rangle\langle 0| \otimes B_j A_i + |1\rangle\langle 1| \otimes A_i B_j$$

$$A[\rho] = \sum_i A_i \rho A_i^\dagger$$

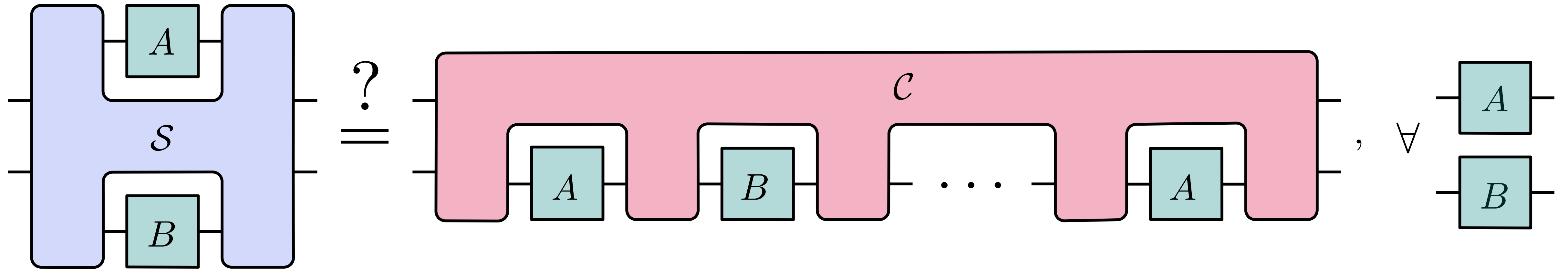
$$B[\rho] = \sum_j B_j \rho B_j^\dagger$$

Simulation

Indefinite causal order



Simulation of indefinite causal order

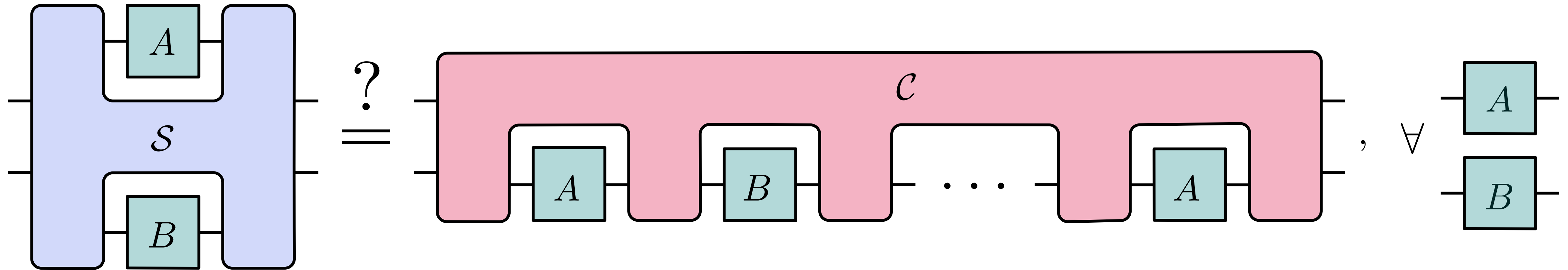


$$\mathcal{C}(A^{\otimes k_A}, B^{\otimes k_B}) = \mathcal{S}(A, B) \quad \forall A, B$$

Simulation of indefinite causal order

Cost: quantum query complexity

(k_A, k_B)

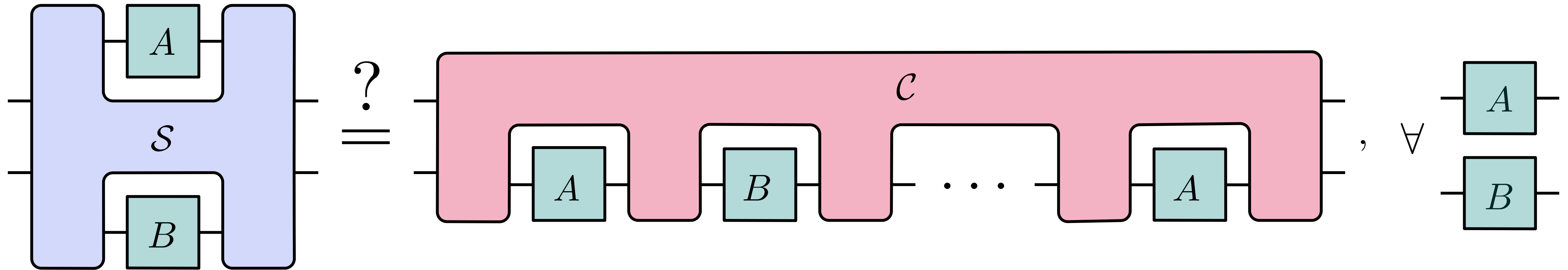


$$\mathcal{C}(A^{\otimes k_A}, B^{\otimes k_B}) = \mathcal{S}(A, B) \quad \forall A, B$$

Rules of the simulation

Cost: quantum query complexity

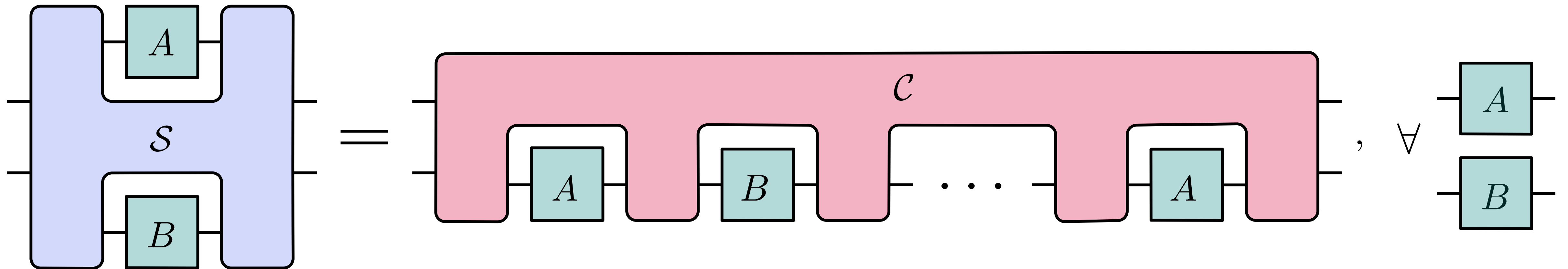
(k_A, k_B)



- Finite number of calls
- Deterministic
- Universal (blackbox inputs)

Simulation of indefinite causal order

Possible strategies for simulation:

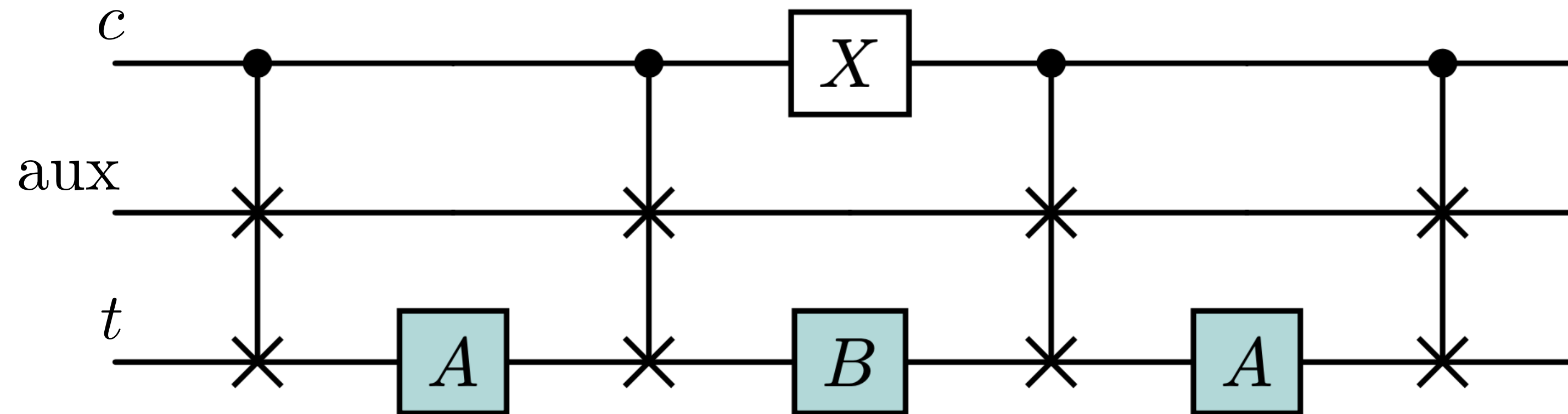


- Infinite number of queries = process tomography
- Approximate / probabilistic simulation [1,2]
- Non-universal (particular inputs) [1]

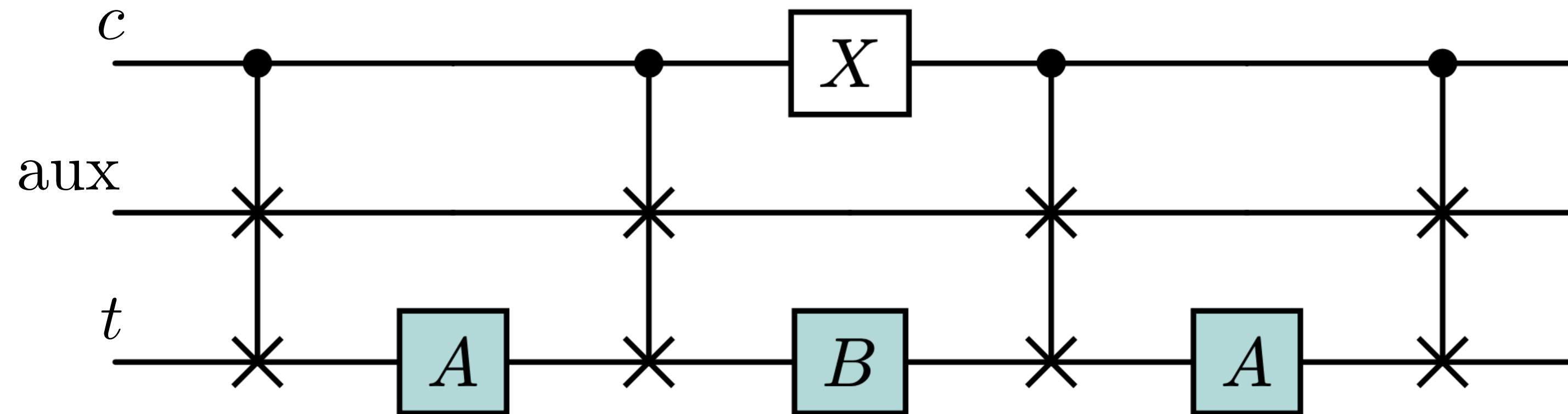
[1] Chiribella, D'Ariano, Perinotti, Valiron, [PRA 88, 022318 \(2013\)](#)

[2] Milz, Pollock, Le, Chiribella, Modi, [NJP 20, 033033 \(2018\)](#), Quintino, Dong, Shimbo, Soeda, Murao, [PRA 100, 062339 \(2019\)](#)

Non-universal simulation: unitary channels

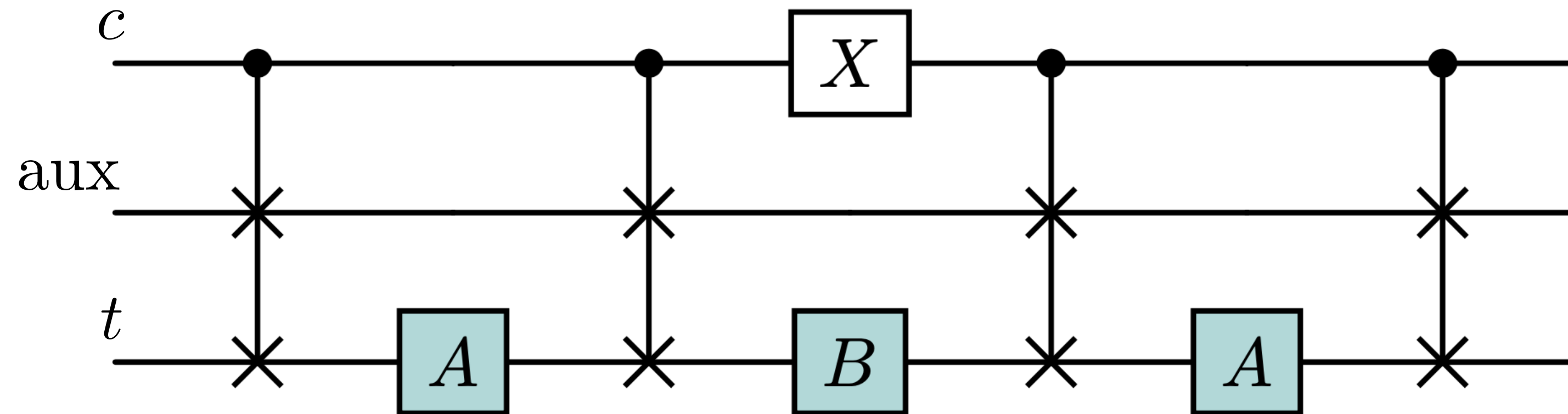


Non-universal simulation: unitary channels



$$C_{iji'} = |0\rangle\langle 0| \otimes A_{i'} \otimes B_j A_i + |1\rangle\langle 1| \otimes A_i \otimes A_{i'} B_j$$

Non-universal simulation: unitary channels



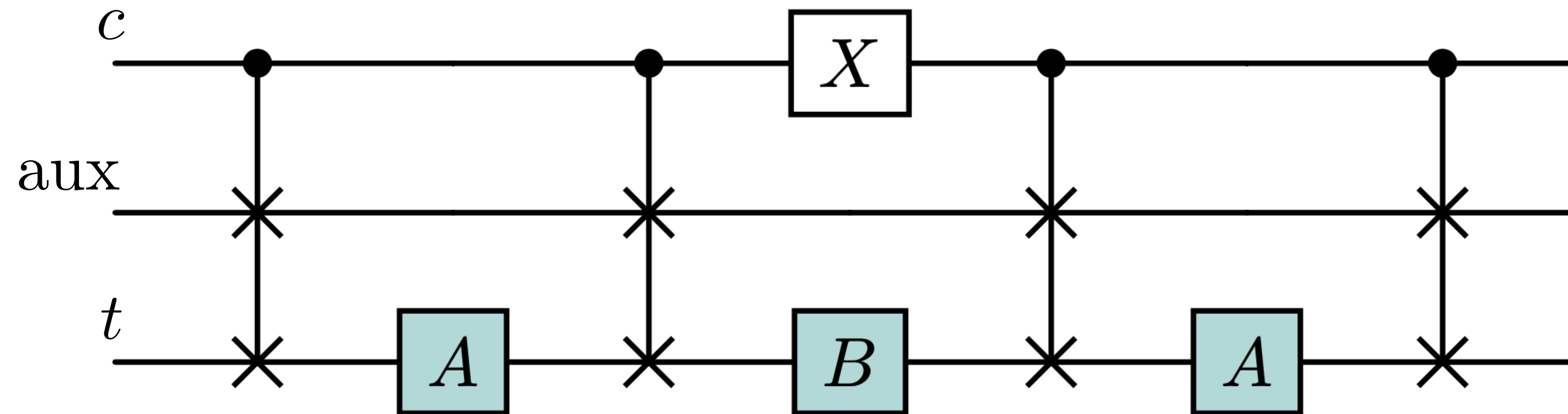
$$C_{ij i'} = |0\rangle\langle 0| \otimes A_{i'} \otimes B_j A_i + |1\rangle\langle 1| \otimes A_i \otimes A_{i'} B_j$$

For unitary channels:

$$A[\rho] = U_A[\rho] = \sum_i A_i \rho A_i^\dagger = A_0 \rho A_0^\dagger$$

$$B[\rho] = U_B[\rho] = \sum_j B_j \rho B_j^\dagger = B_0 \rho B_0^\dagger$$

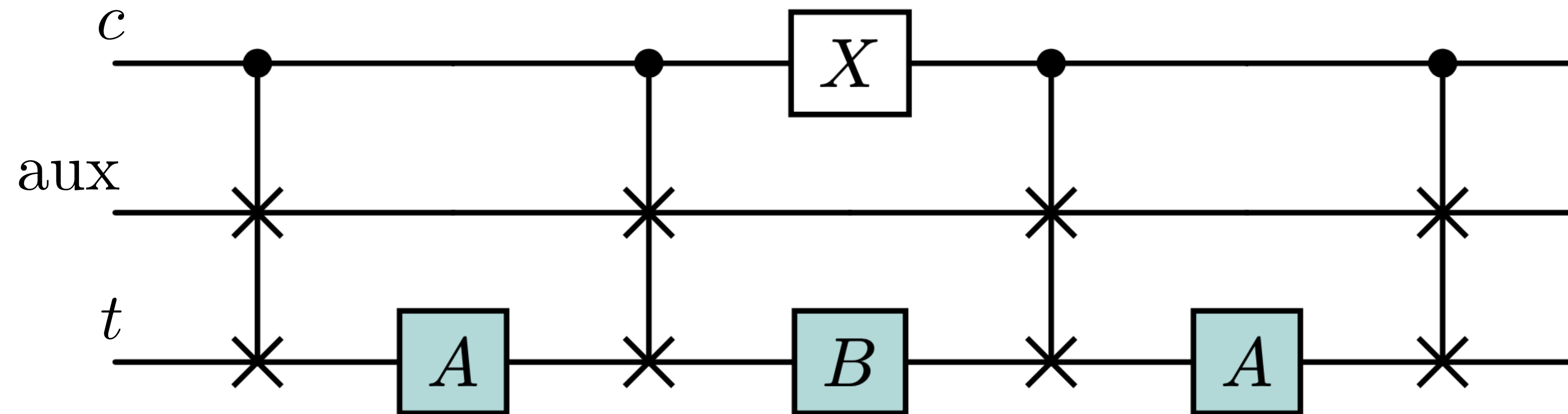
Non-universal simulation: unitary channels



$$C_{000} = |0\rangle\langle 0| \otimes A_0 \otimes B_0 A_0 + |1\rangle\langle 1| \otimes A_0 \otimes A_0 B_0 \quad (\text{quantum circuit})$$

For unitary channels: $S_{00} := |0\rangle\langle 0| \otimes B_0 A_0 + |1\rangle\langle 1| \otimes A_0 B_0 \quad (\text{quantum switch})$

Non-universal simulation: unitary channels

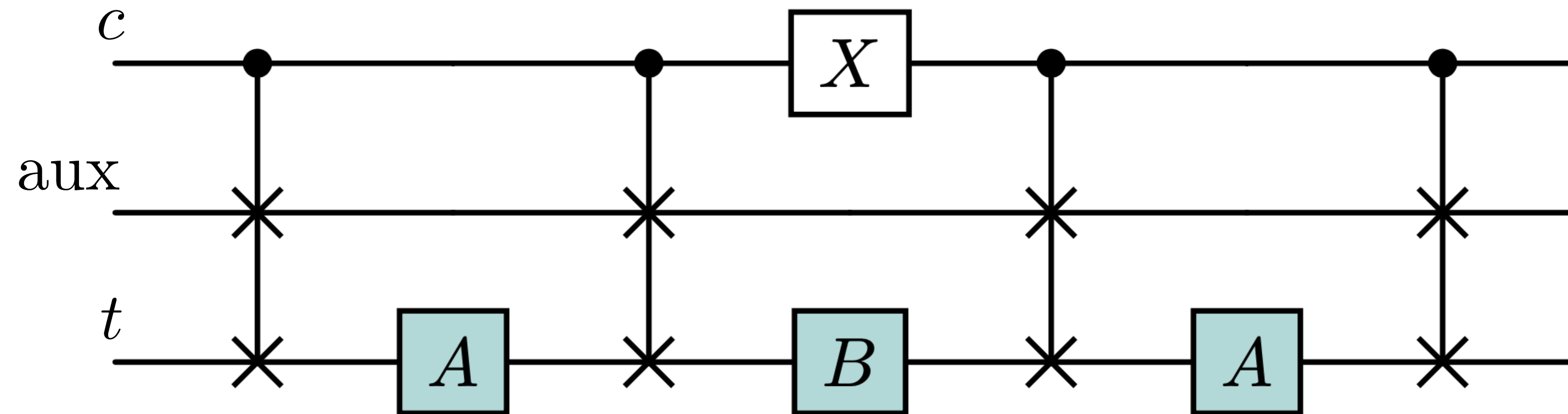


$$C_{000} = |0\rangle\langle 0| \otimes \cancel{A_0} \otimes B_0 A_0 + |1\rangle\langle 1| \otimes \cancel{A_0} \otimes A_0 B_0 \quad (\text{quantum circuit})$$

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Non-universal simulation: unitary channels



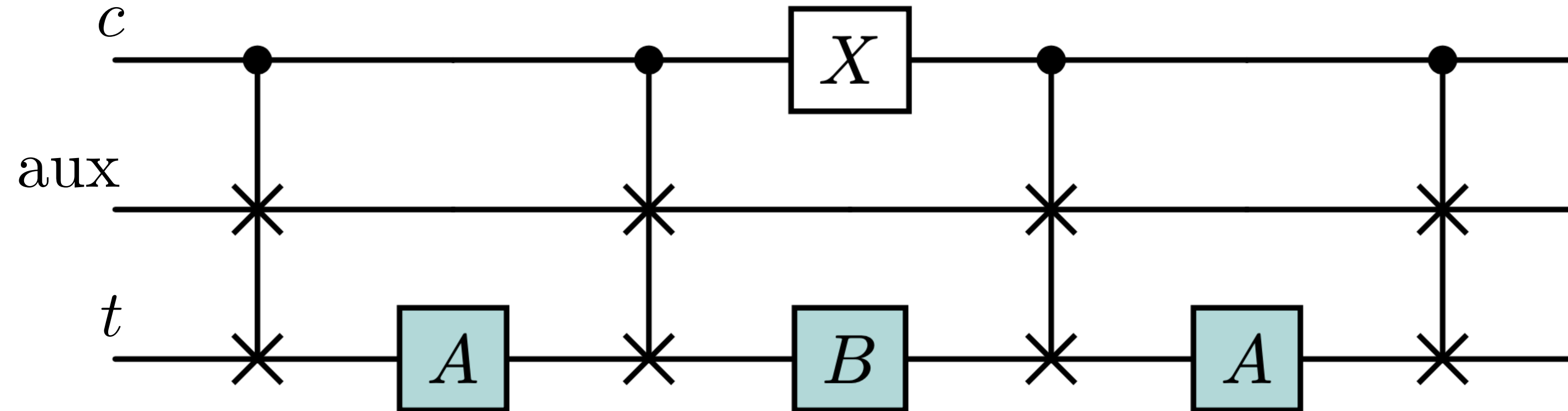
$$C_{ij i'} = |0\rangle\langle 0| \otimes A_{i'} \otimes B_j A_i + |1\rangle\langle 1| \otimes A_i \otimes A_{i'} B_j \quad (\text{quantum circuit})$$

$$\neq$$

For general channels:

$$S_{ij} := |0\rangle\langle 0| \otimes B_j A_i + |1\rangle\langle 1| \otimes A_i B_j \quad (\text{quantum switch})$$

Non-universal simulation: unitary channels

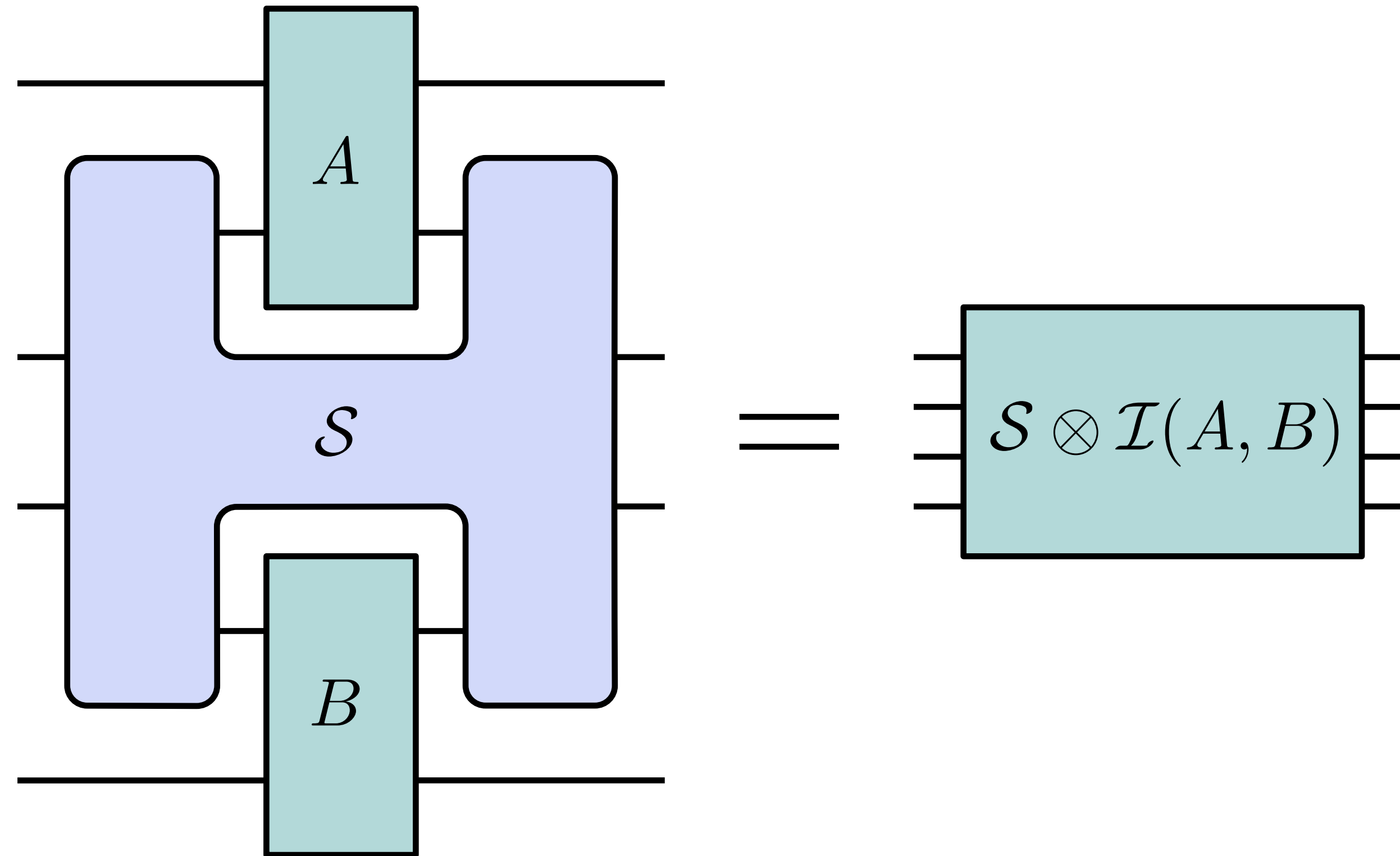


Not a simulation of the quantum switch in general.

But it can be generalized to a certain extent.

Go result: a non-universal simulation

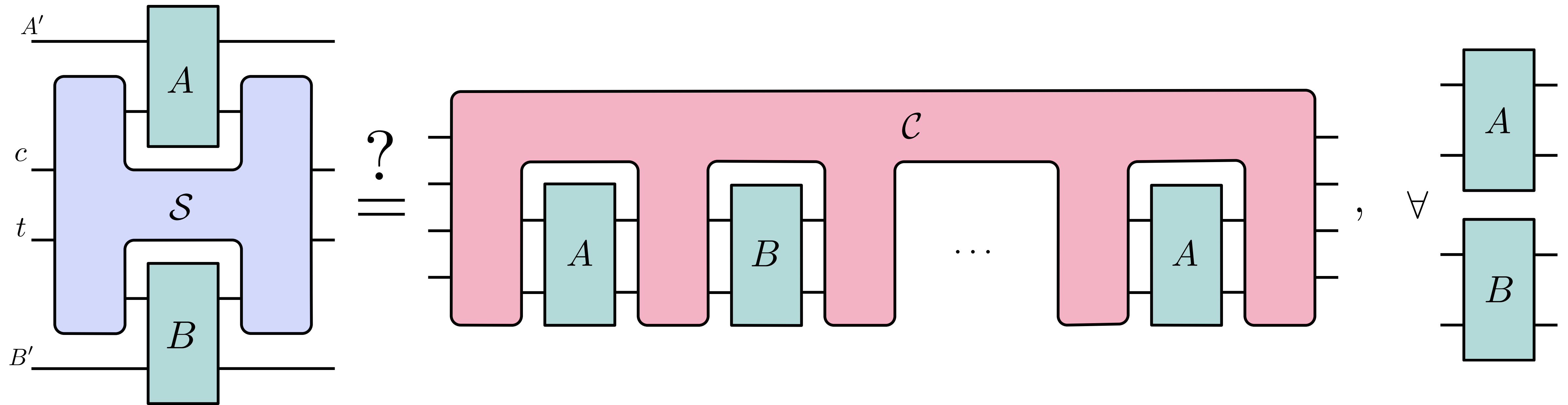
0) The general quantum switch transformation



$$S \otimes \mathcal{I}(A, B)[\sigma_c \otimes \rho_t \otimes \omega_{A'B'}] = \sum_{i,j} S_{ij}(\sigma_c \otimes \rho_t \otimes \omega_{A'B'}) S_{ij}^\dagger$$

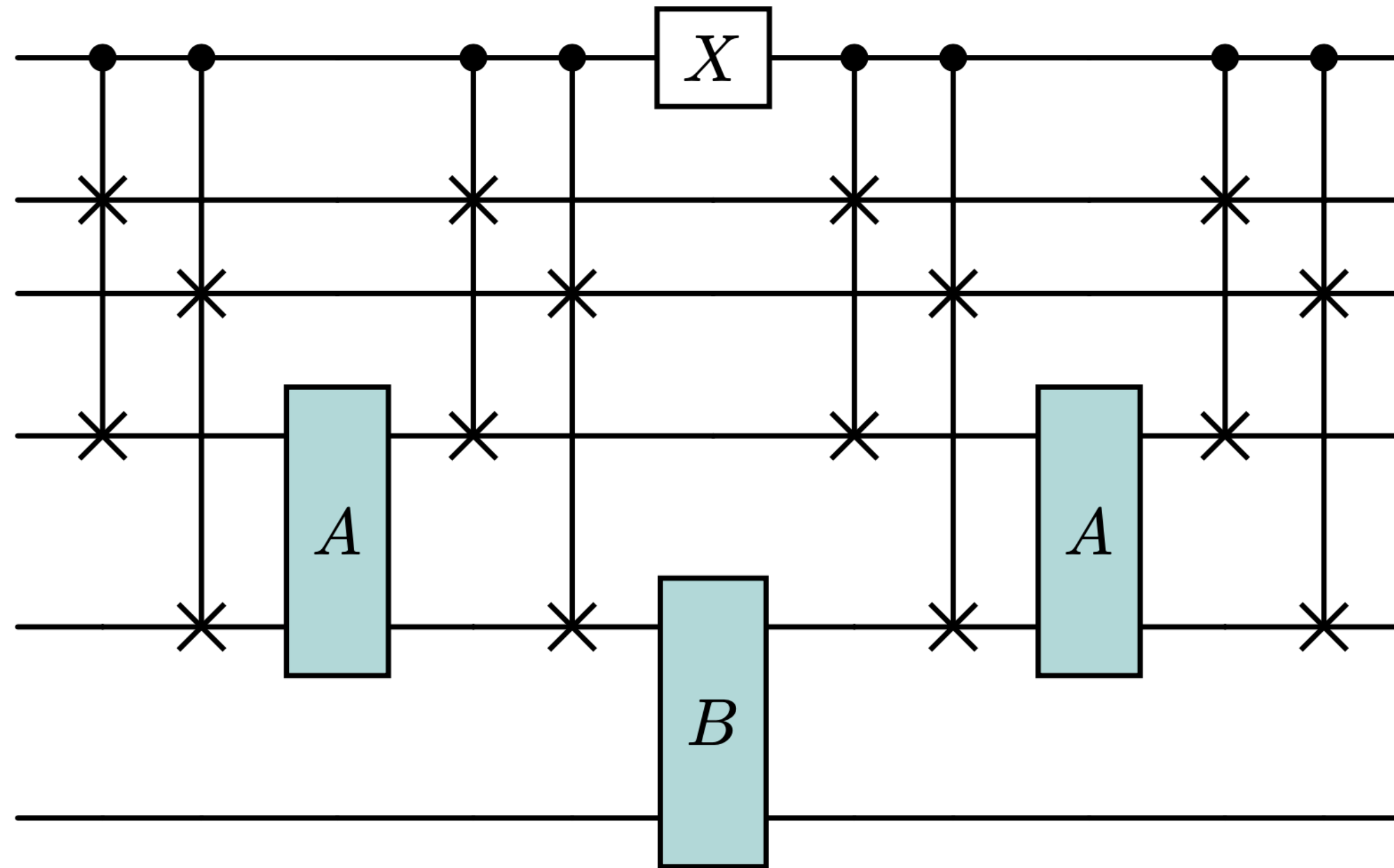
definition implied by linearity

0) The general quantum switch transformation

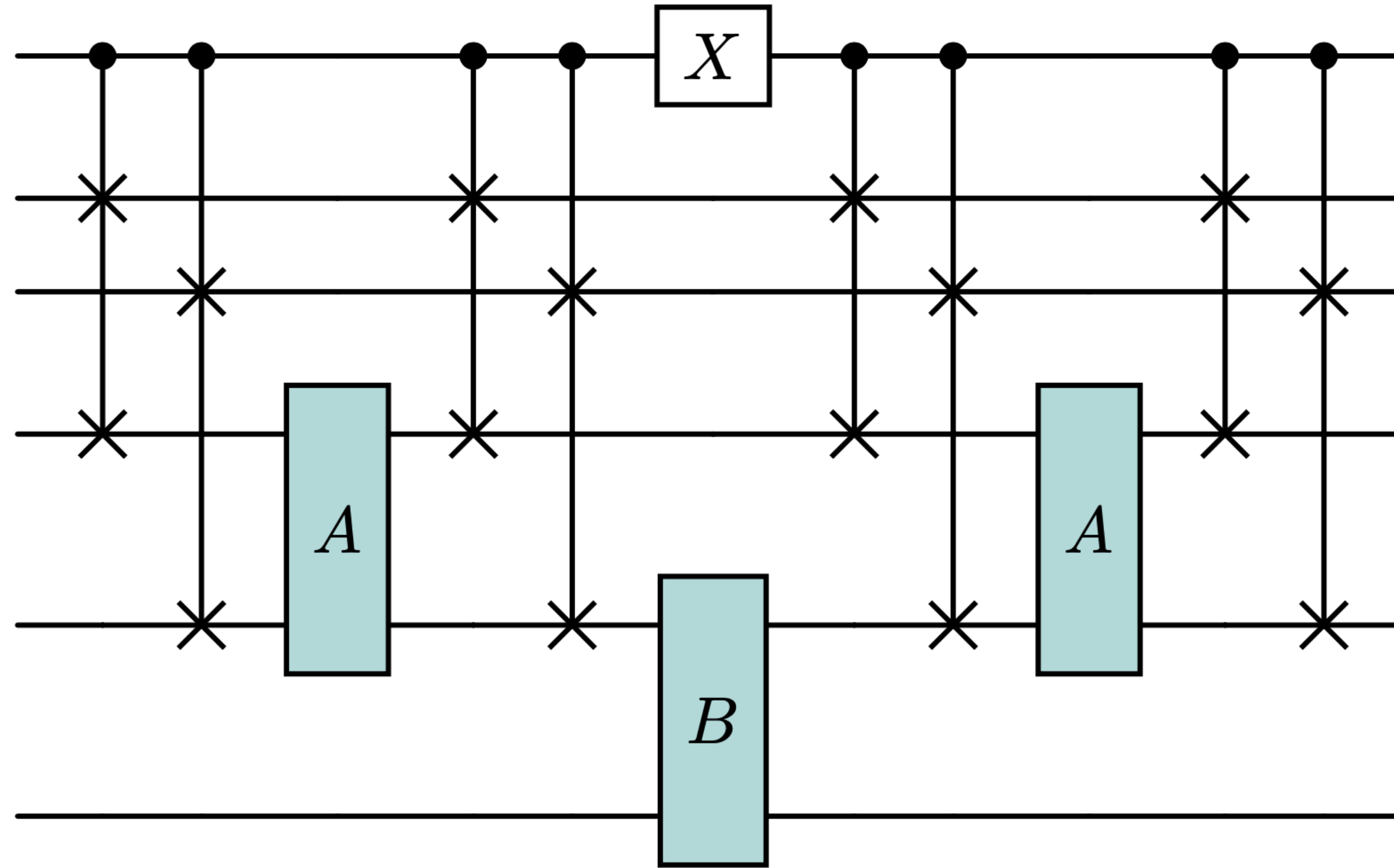


$$\mathcal{C}(A^{\otimes k_A}, B^{\otimes k_B}) = \mathcal{S} \otimes \mathcal{I}(A, B) \quad \forall A, B$$

0) Go: non-universal simulation for extended channels



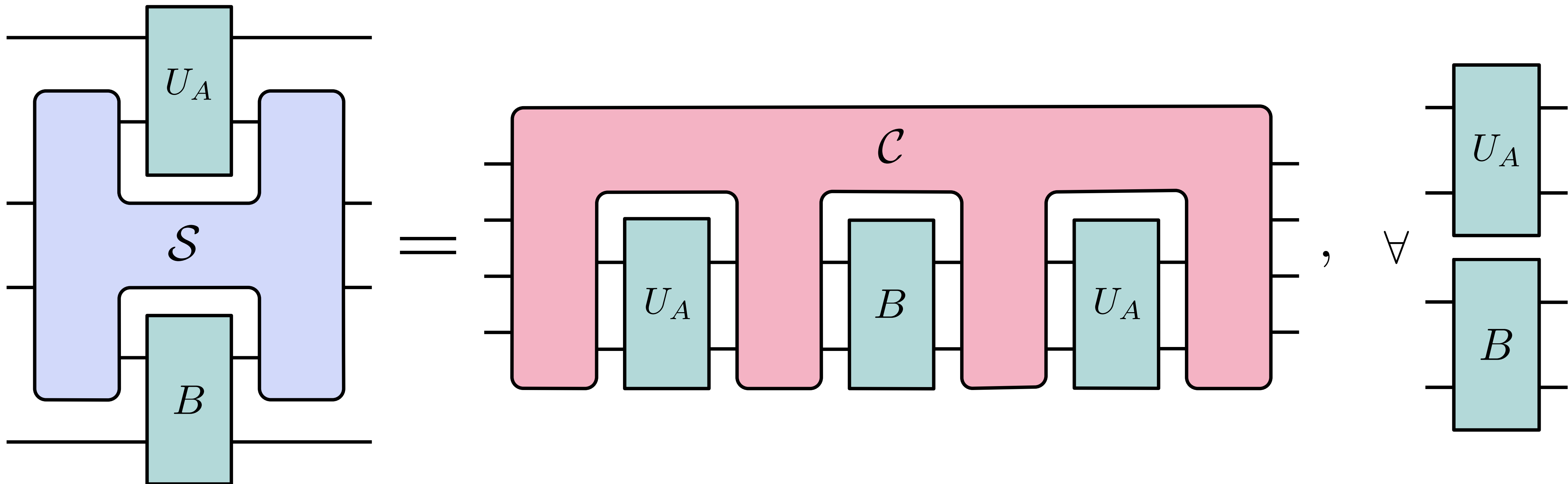
0) Go: non-universal simulation for extended channels



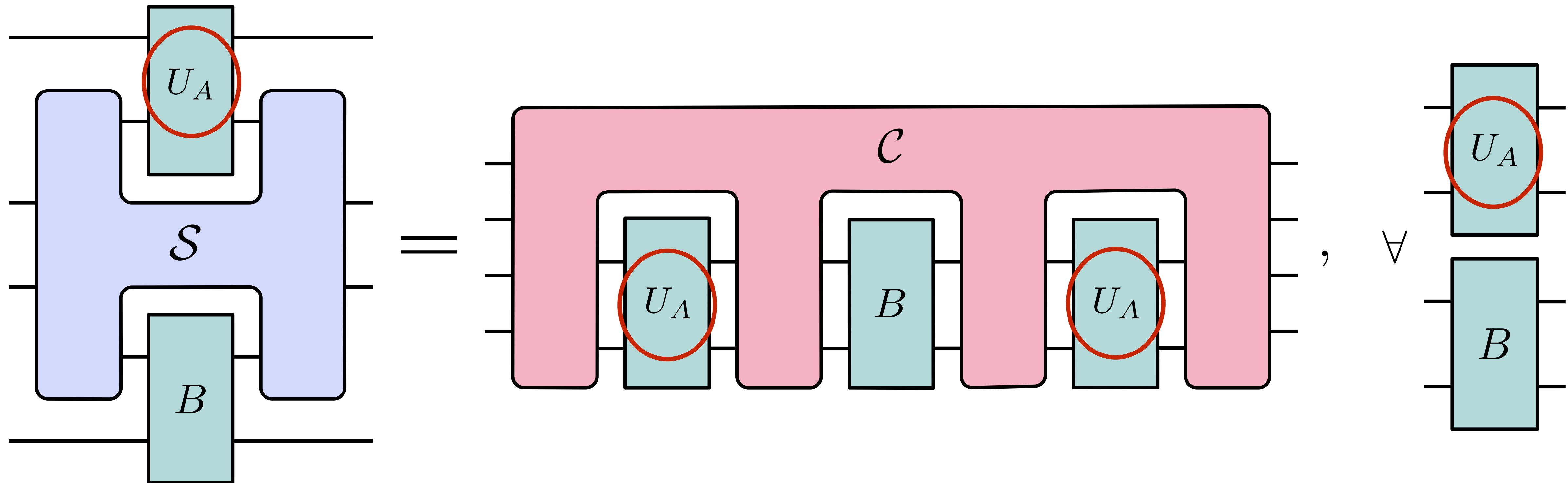
For unitary channels A : $A[\rho] = U_A[\rho] = \sum_i A_i \rho A_i^\dagger = A_0 \rho A_0^\dagger$

For general channels B : $B[\rho] = \sum_j B_j \rho B_j^\dagger$

0) Go: non-universal simulation for extended channels



0) Go: non-universal simulation for extended channels



Without this restriction,
the result does not hold.

No-go results

Main results: no-go theorems

The action of the quantum switch **cannot** be simulated by a quantum circuit that has access to at most (k_A, k_B) calls of the input channels A and B when

$$(k_A, k_B) = (M, 1) \quad \text{for} \quad M < 2^n$$

and

$$(k_A, k_B) \in \{(2, 1), (2, 2), (3, 1)\}$$



number of qubits in
the target system

1) No-go: exponential separation

Theorem 2. *There is no $(k_A + 1)$ -slot higher-order transformation $\mathcal{C}$, described by a quantum circuit with fixed or classically-controlled causal order, that can simulate the quantum switch, i.e., that satisfies*

$$\mathcal{C}(A^{\otimes k_A}, B) = \mathcal{S}(A, B) \quad (8)$$

for all n -qubit mixed unitary channels A and unitary channels B , if

$$k_A \leq \max(2, 2^n - 1). \quad (9)$$

Therefore, such a simulation also does not exist for all n -qubit quantum channels A and B .

1) No-go: exponential separation - proof

- **Step 1:** Take a simulation

(Lemma 2)

$$\mathcal{C}(A^{\otimes k_A}, B^{\otimes k_B}) = \mathcal{S}(A, B)$$

for all mixed unitary channels A, B

1) No-go: exponential separation - proof

- **Step 1:** Take a simulation

(Lemma 2)

$$\mathcal{C}(A^{\otimes k_A}, B^{\otimes k_B}) = \mathcal{S}(A, B)$$

for all mixed unitary channels A, B

Assume

$$C \geq 0 \implies C = \sum_a \left| C^{(a)} \right\rangle \left\langle C^{(a)} \right| \quad \text{and} \quad \text{rank}(S) = 1 \implies S = |S\rangle\langle S|$$

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Then

$$\left| C^{(a)} \right\rangle * \bigotimes_{k=1}^M |U_k\rangle \otimes \bigotimes_{j=1}^N |V_j\rangle \in \text{span}(\{|S\rangle * (|U_k\rangle \otimes |V_l\rangle)\}_{kl}),$$

$$\mathcal{C}(A^{\otimes k_A}, B^{\otimes k_B}) = \mathcal{S}(A, B) \text{ for all mixed unitary channels } A, B$$

$$C = \sum_a \left| C^{(a)} \right\rangle\!\!\left\langle C^{(a)} \right|$$

$$\Downarrow$$

$$\sum_a \left| C^{(a)} \right\rangle\!\!\left\langle C^{(a)} \right| * \left(\bigotimes_{k=1}^M |U_k\rangle\!\langle U_k| \otimes \bigotimes_{l=1}^N |V_l\rangle\!\langle V_l| \right) = \alpha |S\rangle\!\langle S| * \sum_{k=1}^M \sum_{l=1}^N (|U_k\rangle\!\langle U_k| \otimes |V_l\rangle\!\langle V_l|)$$

$$\Downarrow$$

$$\left| C^{(a)} \right\rangle\!\!\left\langle C^{(a)} \right| * \left(\bigotimes_{k=1}^M |U_k\rangle\!\langle U_k| \otimes \bigotimes_{l=1}^N |V_l\rangle\!\langle V_l| \right) \leq \alpha |S\rangle\!\langle S| * \sum_{k=1}^M \sum_{l=1}^N (|U_k\rangle\!\langle U_k| \otimes |V_l\rangle\!\langle V_l|)$$

1) No-go: exponential separation - proof

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$$\iff$$

$$\left| C^{(a)} \right\rangle * (|U_1\rangle \otimes \dots \otimes |U_M\rangle \otimes |V_1\rangle \otimes \dots \otimes |V_N\rangle) = \sum_{k=1}^M \sum_{l=1}^N \xi_{kl}^{(a)}(\{U_i\}_i, \{V_j\}_j) |S\rangle * (|U_k\rangle \otimes |V_l\rangle)$$

1) No-go: exponential separation - proof

- Step 2:

(Lemma 3)

Take

$$\left| C^{(a)} \right\rangle * (|U_1\rangle \otimes \dots \otimes |U_M\rangle \otimes |V_1\rangle \otimes \dots \otimes |V_N\rangle) = \sum_{k=1}^M \sum_{l=1}^N \xi_{kl}^{(a)}(\{U_i\}_i, \{V_j\}_j) |S\rangle * (|U_k\rangle \otimes |V_l\rangle)$$

If the coefficients $\xi_{kl}^{(a)}(\{U_i\}_i, \{V_j\}_j)$ satisfy

1. *independent of U_k and V_l , and*

2. *linear in U_i and V_j for all $i \neq k, j \neq l$,*

Then

$$\left| C^{(a)} \right\rangle = \sum_{k=1}^M \sum_{l=1}^N |S\rangle \otimes \left| \tilde{\xi}_{kl}^{(a)} \right\rangle$$

If the coefficients $\xi_{kl}^{(a)}(\{U_i\}_i, \{V_j\}_j)$ satisfy

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2. *linear in U_i and V_j for all $i \neq k, j \neq l$,*

$$\implies \tilde{\xi}_{kl}(\{U_i\}_{i \neq k}, \{V_j\}_{j \neq l}) = \left\langle \tilde{\xi}_{kl} \left| \left(\bigotimes_{i \neq k} |U_i\rangle \otimes \bigotimes_{j \neq l} |V_j\rangle \right) \right. \right\rangle$$

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$$\left| C^{(a)} \right\rangle * \left[\bigotimes_{i=1}^M |U_i\rangle^{I_i O_i} \otimes \bigotimes_{j=1}^N |V_j\rangle^{I'_j O'_j} \right] = \left[\sum_{k=1}^M \sum_{l=1}^N \left| \tilde{\xi}_{kl} \right\rangle \otimes |S\rangle \right] * \left[\bigotimes_{i=1}^M |U_i\rangle^{I_i O_i} \otimes \bigotimes_{j=1}^N |V_j\rangle^{I'_j O'_j} \right]$$

1) No-go: exponential separation - proof

- Step 2:

(Lemma 3)

Take

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1) No-go: exponential separation - proof

- Step 3:

(Lemma 4)

Take

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If

Alice's copies

$$M < 4^n/2 + 2 \text{ and } N = 1$$

Bob's copies

Then there exist coefficients $\xi_{kl}^{(a)}(\{U_i\}_i, \{V_j\}_j)$ that are

1. *independent of U_k and V_l , and*

2. *linear in U_i and V_j for all $i \neq k, j \neq l$,*

1) No-go: exponential separation - proof

- Step 4:
(Lemma 5)

Take

$$\left| C^{(a)} \right\rangle = \sum_{k=1}^M \sum_{l=1}^N |S\rangle \otimes \left| \tilde{\xi}_{kl}^{(a)} \right\rangle$$

If

$|S\rangle$ is the Choi vector of the quantum switch

and

$$\max\{M, N\} \leq \max\{2, d - 1\}$$

Then

$$C = \sum_a \left| C^{(a)} \right\rangle \left\langle C^{(a)} \right| \notin \text{QC-CC}$$

1) No-go: exponential separation

Theorem 2. *There is no $(k_A + 1)$ -slot higher-order transformation $\mathcal{C}$, described by a quantum circuit with fixed or classically-controlled causal order, that can simulate the quantum switch, i.e., that satisfies*

$$\mathcal{C}(A^{\otimes k_A}, B) = \mathcal{S}(A, B) \quad (8)$$

for all n -qubit mixed unitary channels A and unitary channels B , if

$$k_A \leq \max(2, 2^n - 1). \quad (9)$$

Therefore, such a simulation also does not exist for all n -qubit quantum channels A and B .

2) No-go: probabilistic simulation

Theorem 3. *There is no quantum circuit that can deterministically simulate the action of the quantum switch on all quantum channels when*

$$(k_A, k_B) = (2, 2), \quad (13)$$

where k_A is the number of calls of input channel A and k_B is the number of calls of input channel B.

Moreover, even for a restricted simulation—i.e., for fixed input systems and discarded output target system—the action on the quantum switch on general qubit channels, when

$$(k_A, k_B) \in \{(1, 1), (2, 1), (2, 2), (3, 1)\}, \quad (14)$$

can be simulated with at most a probability $p < 1$, with upper bounds given in Table I.

(k_A, k_B)	order	probability
(1, 1)	AB	$p < \frac{4001}{10000}$
(2, 1)	AAB	$p < \frac{5715}{10000}$
	ABA	$p < \frac{4919}{10000}$
	BAA	$p < \frac{5001}{10000}$
(2, 2)	AABB	$p < \frac{8307}{10000}$
	ABAB	$p < \frac{8484}{10000}$
	ABBA	$p < \frac{8695}{10000}$
(3, 1)	AAAB	$p < \frac{8373}{10000}$
	AABA	$p < \frac{6909}{10000}$
	ABAA	$p < \frac{7597}{10000}$
	BAAA	$p < \frac{6845}{10000}$

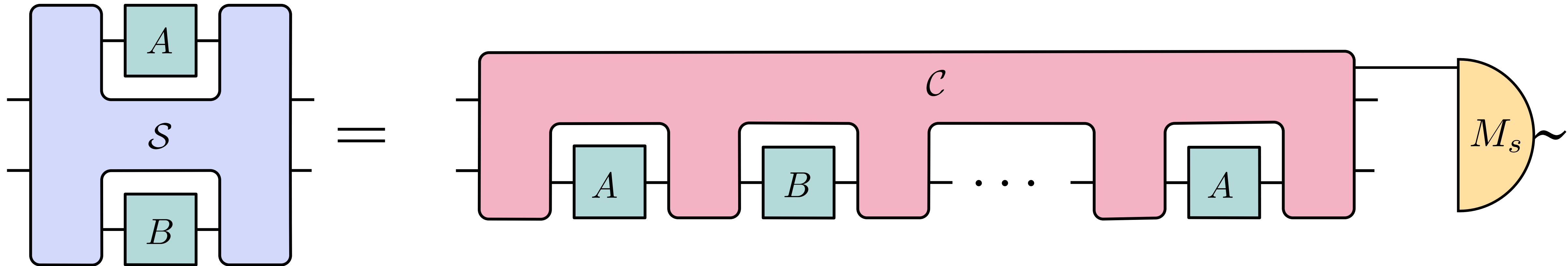
2) No-go: probabilistic simulation - proof

(identical channels)

k	order	probability
2	AA	$p < \frac{4001}{10000}$
3	AAA	$p < \frac{6534}{10000}$
4	AAAA	$p = 1 \text{ (*)}$

2) No-go: probabilistic simulation - proof

Technique: Probabilistic simulation

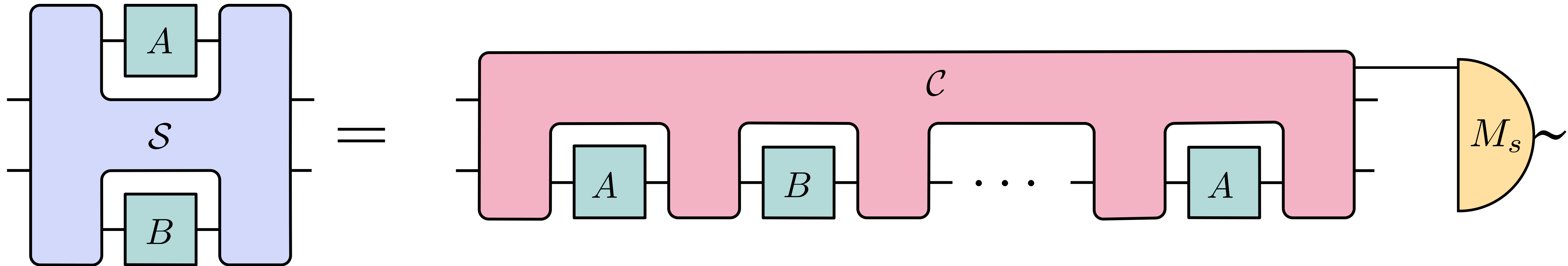


$$\mathcal{C}_s(A^{\otimes k_A}, B^{\otimes k_B}) = p \mathcal{S}(A, B) \quad \forall A, B$$

$$\mathcal{C} = \mathcal{C}_s + \mathcal{C}_f$$

2) No-go: probabilistic simulation - proof

Technique: Probabilistic simulation



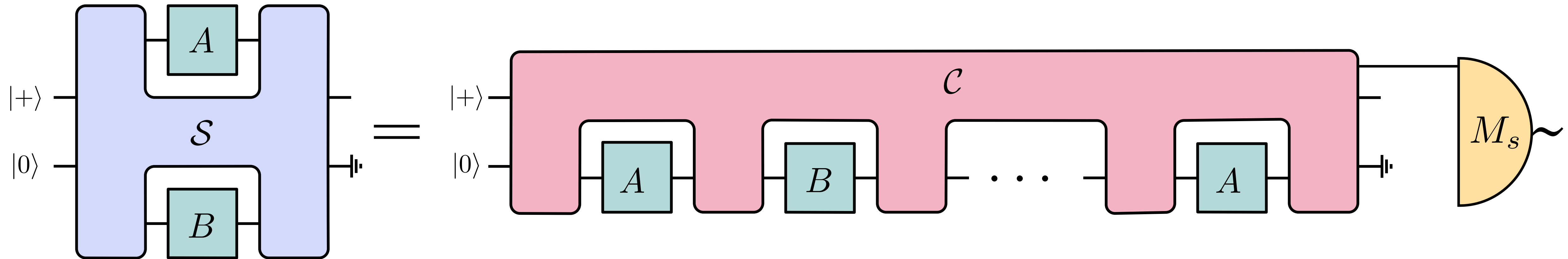
$$\mathcal{C}_s(A^{\otimes k_A}, B^{\otimes k_B}) = \textcircled{p} \mathcal{S}(A, B) \quad \forall A, B$$

$$\mathcal{C} = \mathcal{C}_s + \mathcal{C}_f$$

$\max p < 1 \implies$ impossible simulation

2) No-go: probabilistic simulation - proof

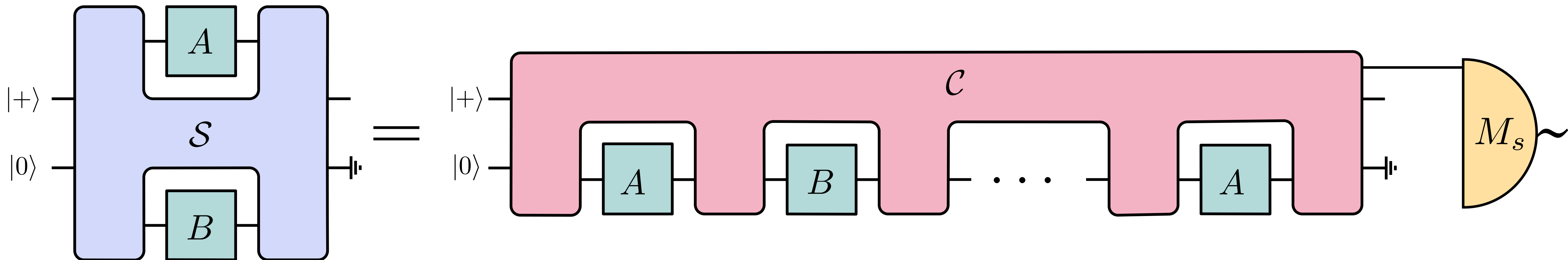
Technique: Restricted simulation



$$\text{tr}_{t_O} \left(\mathcal{C}(A^{\otimes k_A}, B^{\otimes k_B})[|+\rangle\langle+| \otimes |0\rangle\langle 0|] \right) = \text{tr}_{t_O} \left(\mathcal{S}(A, B)[|+\rangle\langle+| \otimes |0\rangle\langle 0|] \right) \quad \forall A, B$$

2) No-go: probabilistic simulation - proof

Technique: Restricted simulation

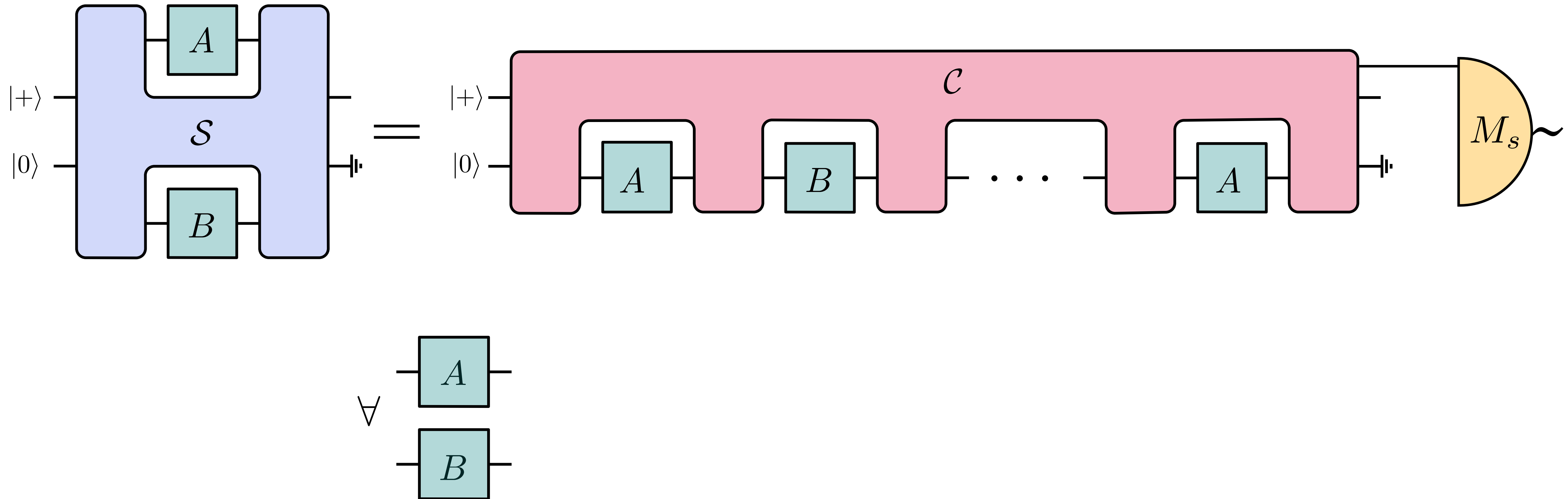


$$\text{tr}_{t_O} \left(\mathcal{C}(A^{\otimes k_A}, B^{\otimes k_B})[|+\rangle\langle+| \otimes |0\rangle\langle 0|] \right) = \text{tr}_{t_O} \left(\mathcal{S}(A, B)[|+\rangle\langle+| \otimes |0\rangle\langle 0|] \right) \quad \forall A, B$$

restricted simulation impossible $\implies$ general simulation impossible

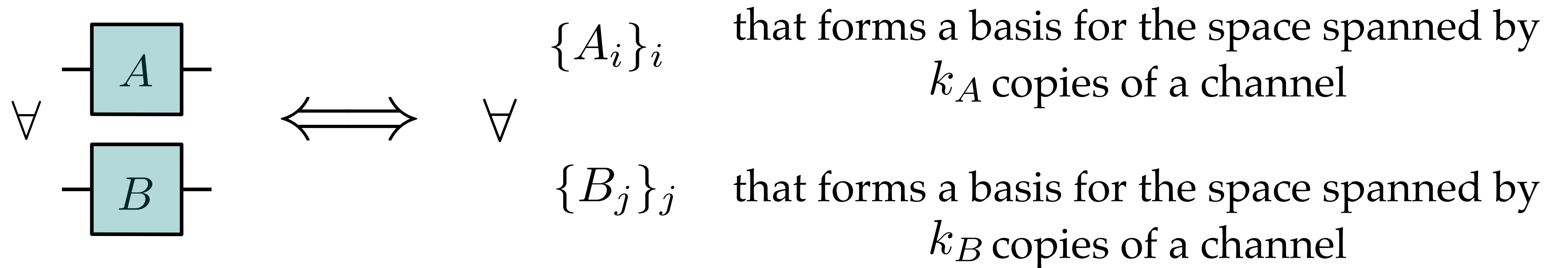
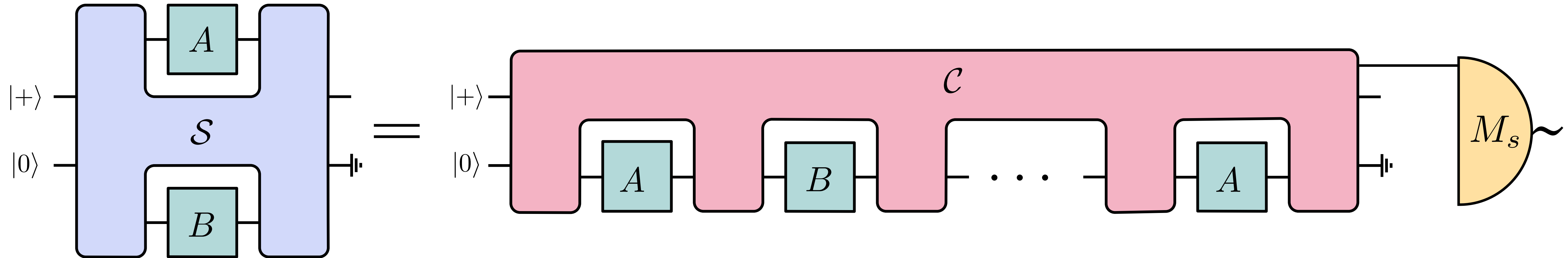
2) No-go: probabilistic simulation - proof

Technique: Input basis



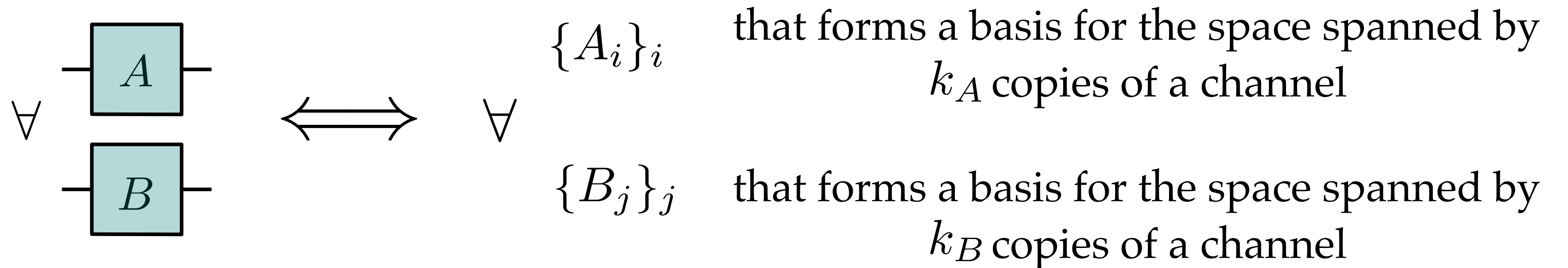
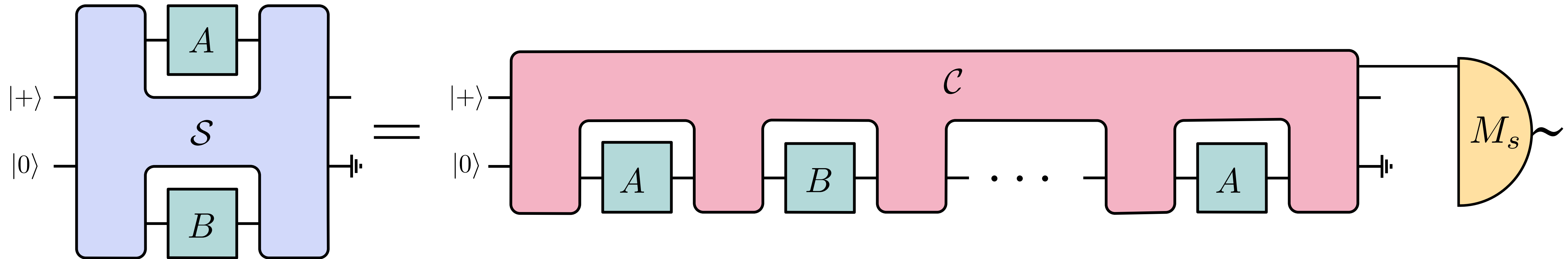
2) No-go: probabilistic simulation - proof

Technique: Input basis

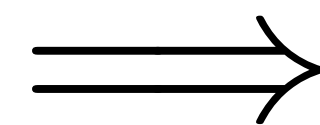


2) No-go: probabilistic simulation - proof

Technique: Input basis



Maximum probability of success for
input channel basis



Maximum probability of success for
all channels

2) No-go: probabilistic simulation - proof

Technique: Semidefinite programming

given $\{J_i^A\}_i, \{J_j^B\}_j, k_A, k_B$

max p

s.t. $C_s * [(J_i^A)^{\otimes k_A} \otimes (J_j^B)^{\otimes k_B}] = p S * (J_i^A \otimes J_j^B) \quad \forall i, j$

$C_s \geq 0, \quad C - C_s \geq 0,$

$\mathbb{P}(C) = C, \quad \text{tr}(C) = d_{c_I} d_{t_I} d_{A_O}^{k_A} d_{B_O}^{k_B},$

given $\{J_i^A\}_i, \{J_j^B\}_j, k_A, k_B$

min $\frac{1}{d_{A_I}^{k_A} d_{B_I}^{k_B} d_{c_O} d_{t_O}} \text{tr}(\Gamma)$

s.t. $\sum_{i,j} \text{tr}[R_{ij} (S * (J_i^A \otimes J_j^B))] = 1$

$\Gamma - \sum_{i,j} R_{ij} \otimes [(J_i^A)^{\otimes k_A} \otimes (J_j^B)^{\otimes k_B}]^T \geq 0$

$\Gamma \geq 0, \quad \bar{\mathbb{P}}(\Gamma) = \Gamma,$

any feasible point that yields some
 $p < 1$
constitutes a valid upper bound

2) No-go: probabilistic simulation - proof

Technique: Computer-assisted proofs

Algorithm:

1. Construct symbolic non-floating point operators Γ^{sym} and R_{ij}^{sym} by truncating them and obtaining a symbolic operator with only rational numbers.
2. Force the operators Γ^{sym} and R_{ij}^{sym} to be self-adjoint by making use of the expression $(M + M^\dagger)/2$, which is self-adjoint for any M .
3. Evaluate $t^{\text{sym}} := \sum_{i,j} \text{tr}[R_{ij}^{\text{sym}} (S * (J_i^A \otimes J_j^B))]$, where S, J_i^A , and J_j^B are also symbolic operators. Define $R_{ij}^{\text{ok}} := R_{ij}^{\text{sym}}/t^{\text{sym}}$ for all i, j .
4. Project Γ^{sym} onto the appropriate subspace to obtain $\bar{\mathbb{P}}(\Gamma^{\text{sym}})$.
5. Find $\eta \in \mathbb{R}$ such that $\Gamma^{\text{ok}} := \bar{\mathbb{P}}(\Gamma^{\text{sym}}) + \eta \mathbb{1} \geq 0$ and $\Gamma^{\text{ok}} - \sum_{i,j} R_{ij}^{\text{ok}} \otimes (J_i^{A \otimes k_A} \otimes J_j^{B \otimes k_B})^T \geq 0$.
6. Output the quantity $\text{tr}(\Gamma^{\text{ok}})/d_{c_I} d_{t_I} d_{A_O}^{k_A} d_{B_O}^{k_B}$, which is a rigorous upper bound of the primal problem.

(k_A, k_B)	order	probability
(1, 1)	AB	$p < \frac{4001}{10000}$
(2, 1)	AAB	$p < \frac{5715}{10000}$
	ABA	$p < \frac{4919}{10000}$
	BAA	$p < \frac{5001}{10000}$
(2, 2)	AABB	$p < \frac{8307}{10000}$
	ABAB	$p < \frac{8484}{10000}$
	ABBA	$p < \frac{8695}{10000}$
(3, 1)	AAAB	$p < \frac{8373}{10000}$
	AABA	$p < \frac{6909}{10000}$
	ABAA	$p < \frac{7597}{10000}$
	BAAA	$p < \frac{6845}{10000}$

Restricted qubit simulations

(identical channels)

k	order	probability
2	AA	$p < \frac{4001}{10000}$
3	AAA	$p < \frac{6534}{10000}$
4	AAAA	$p = 1 \text{ (*)}$

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Simulation with full output: $p < 1$

(k_A, k_B)	order (unitary only)	probability
(1, 1)	AB	$p \approx 0.400$
(2, 1)	AAB	$p \approx 0.596$
	ABA	$p = 1$
	BAA	$p \approx 0.607$
(2, 2)	AABB	$p = 1 \text{ (*)}$
	ABAB	$p = 1$
	ABBA	$p = 1$
(3, 1)	AAAB	$p \approx 0.708$
	AABA	$p = 1$
	ABAA	$p = 1$
	BAAA	$p = 1 \text{ (*)}$

Restricted qubit unitary simulations

Restricted qubit unitary simulations

(k_A, k_B)	order (unitary only)	probability
(1, 1)	AB	$p \approx 0.400$
(2, 1)	AAB	$p \approx 0.596$
	ABA	$p = 1$
	BAA	$p \approx 0.607$
(2, 2)	AABB	$p = 1^{(*)}$
	ABAB	$p = 1$
	ABBA	$p = 1$
(3, 1)	AAAB	$p \approx 0.708$
	AABA	$p = 1$
	ABAA	$p = 1$
	BAAA	$p = 1^{(*)}$

Simulation with full output:

$p < 1$

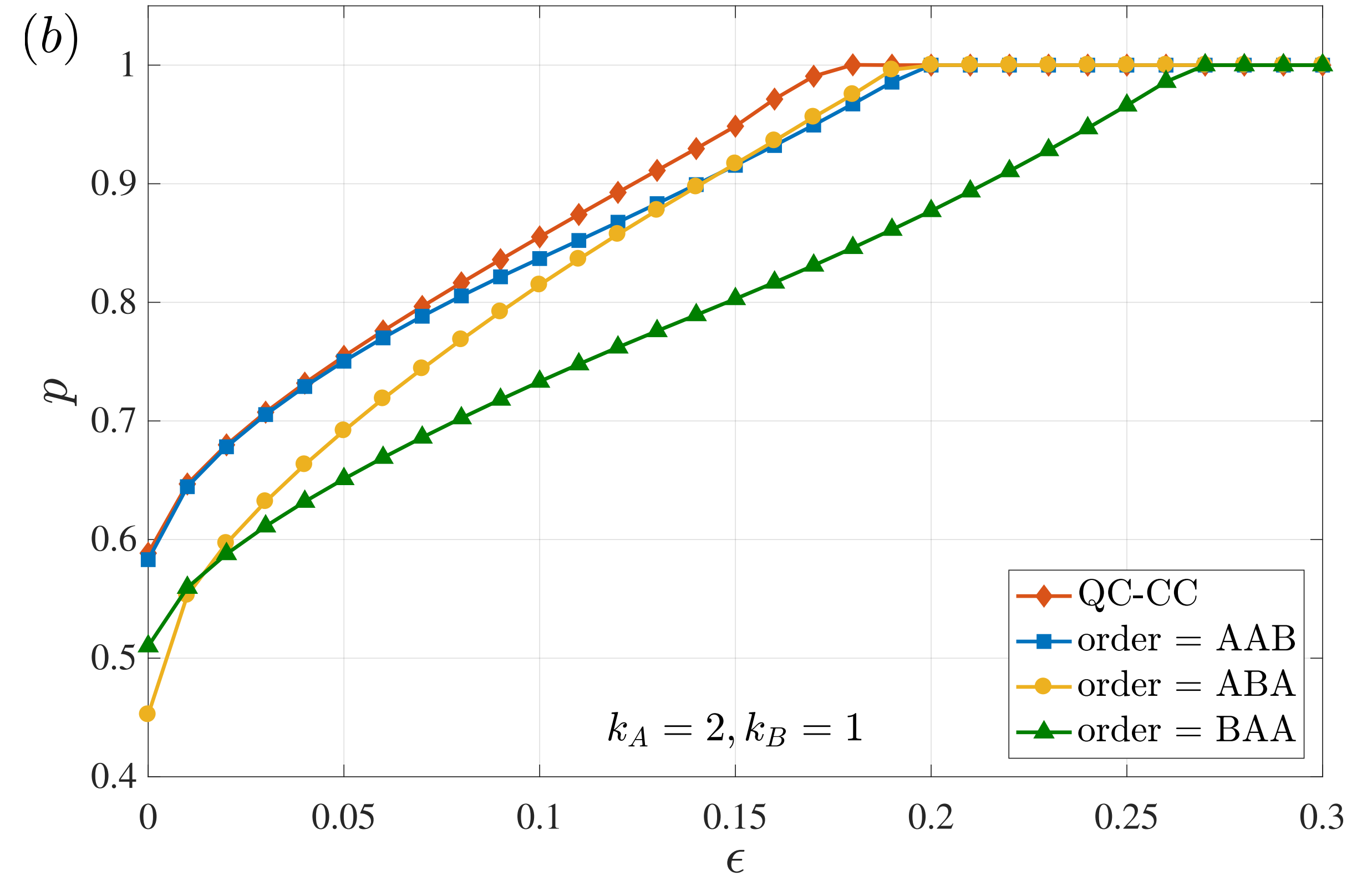
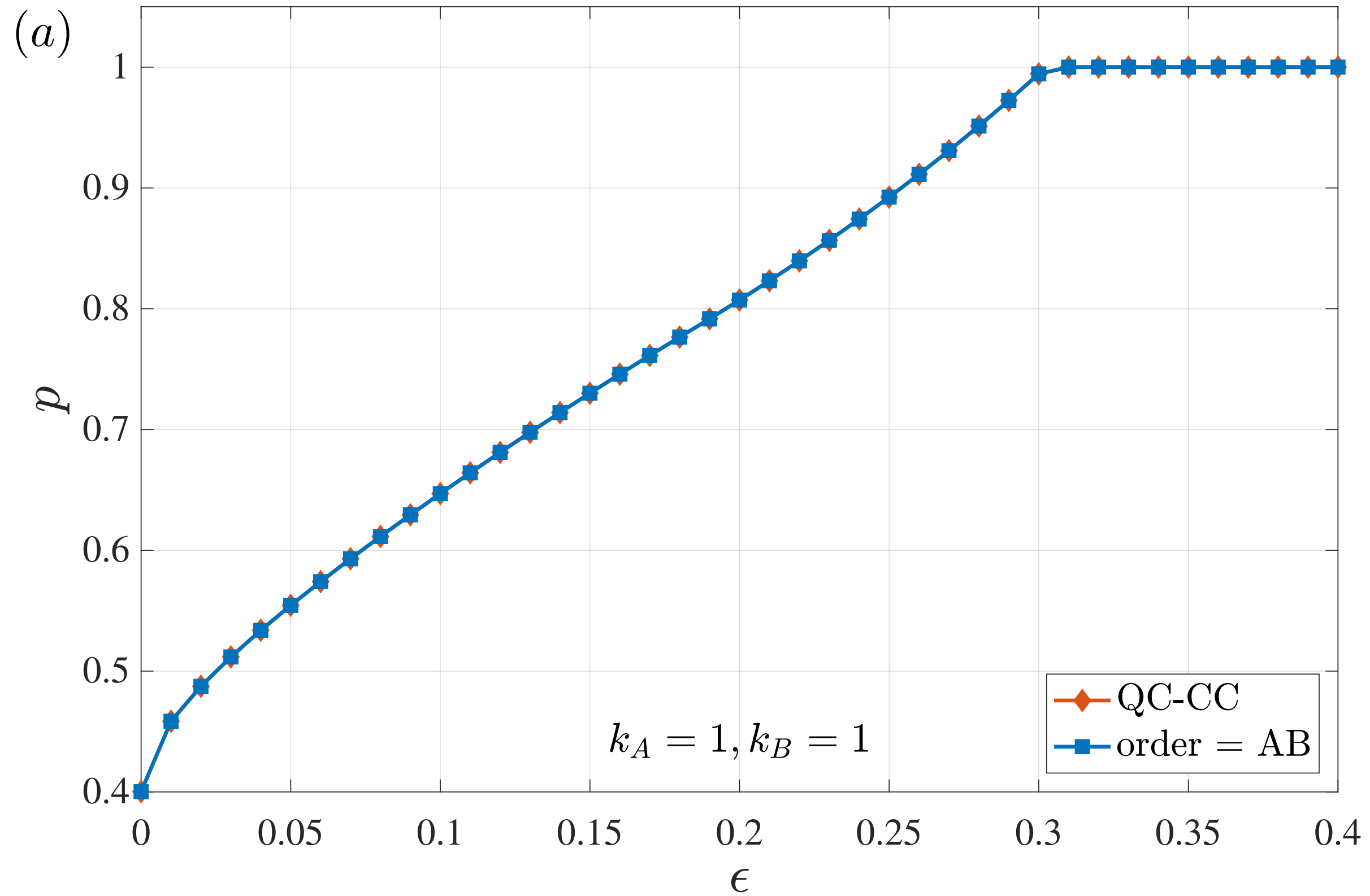
$p < 1$

3) No-go: probabilistic approximate simulation

$$F(\mathcal{S}, \tilde{\mathcal{S}}) \geq 1 - \epsilon$$

3) No-go: probabilistic approximate simulation

$$F(\mathcal{S}, \tilde{\mathcal{S}}) \geq 1 - \epsilon$$



Conclusions

- Higher-order quantum operations

- **No-go theorem:** the action of the quantum switch cannot be simulated by a quantum circuit that has access to $(2,1)$, $(2,2)$, or $(3,1)$ queries of the input channels
- **Robustness:** probability $p < 1$ for $\epsilon > 0$
- **Exponential separation:** $(k_A, k_B) = (M, 1)$ for $M < 2^n$

- Open questions

- Can the quantum switch be simulated with a **finite** number of queries?
- If yes, how is the **scaling**?
- What about **other** higher-order transformations with indefinite causal order?

Thank you!

Extra

Basis

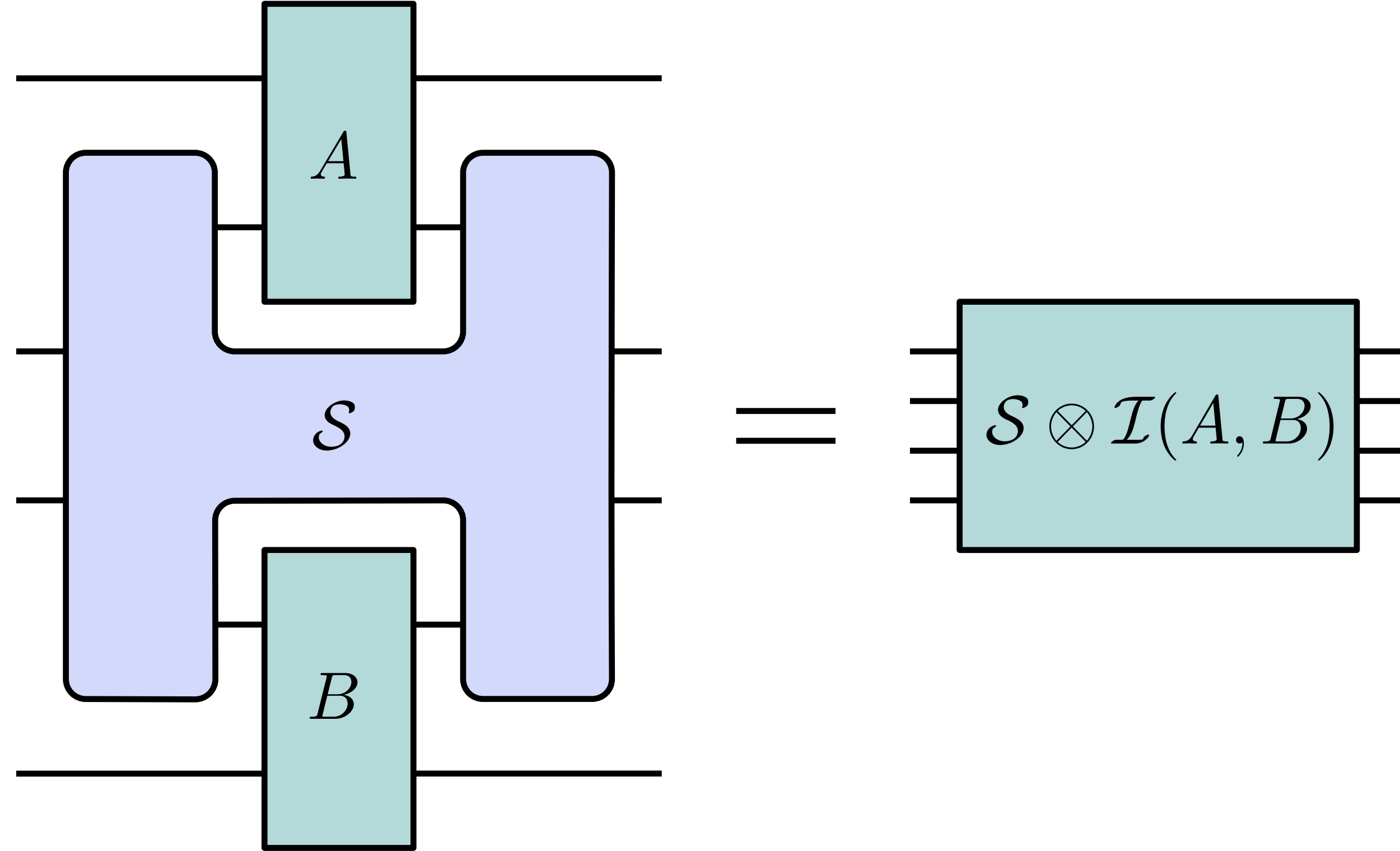
$$\mathcal{B}_1 := \left\{ \begin{aligned} &\mathbb{1} \otimes \frac{\mathbb{1}}{2}, \\ &\mathbb{1} \otimes \frac{\mathbb{1}}{2} + \mathbb{1} \otimes \sigma_i, \\ &\mathbb{1} \otimes \frac{\mathbb{1}}{2} + \sigma_j \otimes \sigma_k \end{aligned} \right\}_{i,j,k}$$

$$\mathcal{B}_2 := \left\{ \begin{aligned} &\left(\mathbb{1} \otimes \frac{\mathbb{1}}{2} \right)^{\otimes 2}, \\ &\left(\rho_i \pm \sigma_j^I \otimes \sigma_k^O \right)^{\otimes 2}, \\ &\left(\rho_l + \sigma_m^I \otimes \sigma_n^O + \sigma_p^I \otimes \sigma_q^O \right)^{\otimes 2} \end{aligned} \right\}_{i,j,k,l,m,n,p,q}$$

$$\mathcal{B}_3 := \left\{ \left(\rho_i \pm \sigma_j^I \otimes \sigma_k^O + \sigma_l^I \otimes \sigma_m^O \right)^{\otimes 3} \right\}_{i,j,k,l,m}$$

$$\left\{ \left(\mathbb{1} \otimes \frac{\mathbb{1}}{2} + \sum_i \alpha_{i|l} \mathbb{1} \otimes \sigma_i + \sum_{ij} \gamma_{ij|l} \sigma_i \otimes \sigma_j \right)^{\otimes k} \right\}_l$$

The general quantum switch transformation

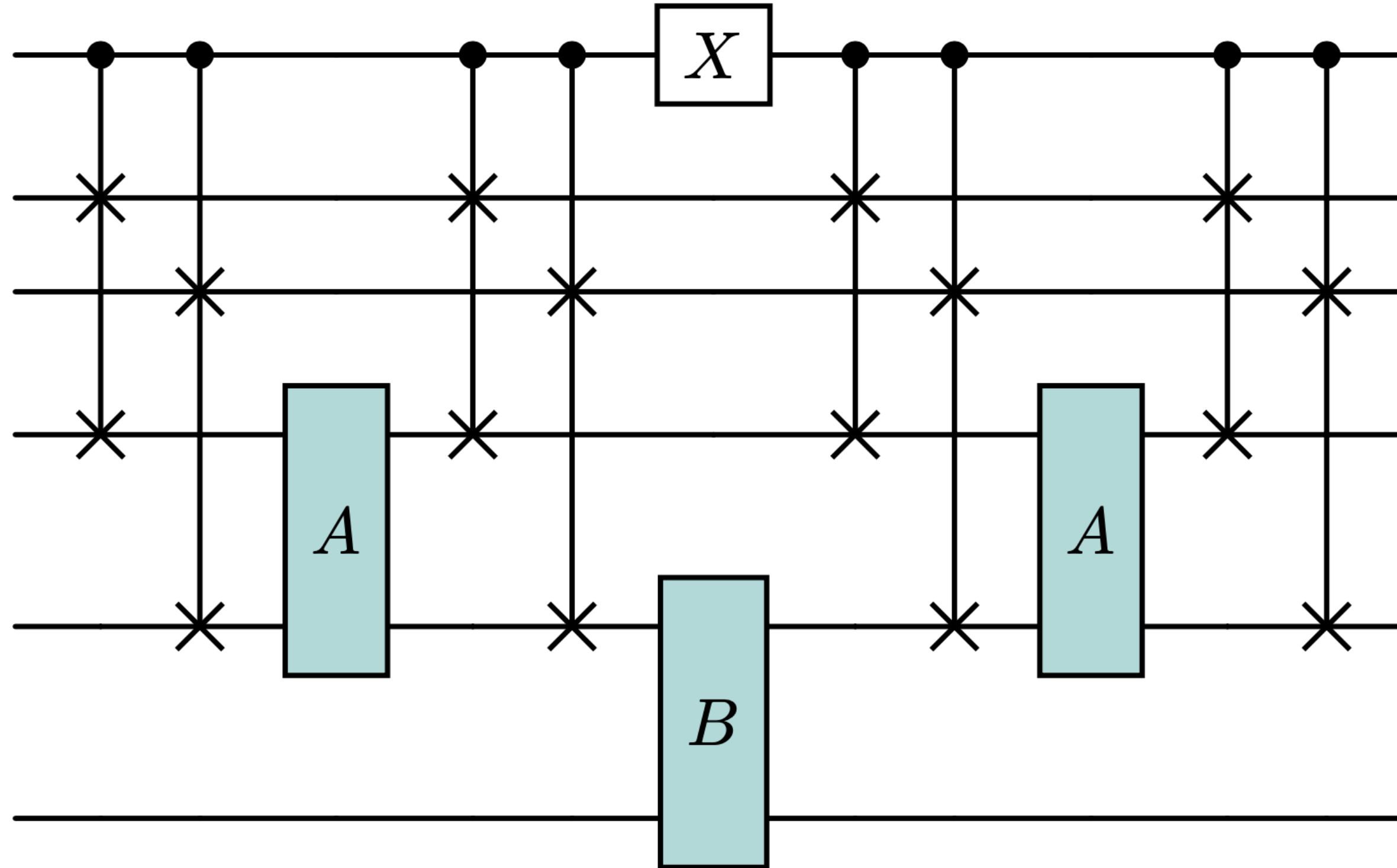


$$\mathcal{S} \otimes \mathcal{I}(A, B)[\sigma_c \otimes \rho_t \otimes \omega_{A'B'}] = \sum_{i,j} S_{ij}(\sigma_c \otimes \rho_t \otimes \omega_{A'B'}) S_{ij}^\dagger$$

$$S_{ij} = |0\rangle\langle 0| \otimes \sum_{ab} \alpha_{a|i} \beta_{b|j} B(b)A(a) \otimes A'(a) \otimes B'(b) + |1\rangle\langle 1| \otimes \sum_{ab} \alpha_{a|i} \beta_{b|j} A(a)B(b) \otimes A'(a) \otimes B'(b)$$

$$A_i = \sum_a \alpha_{a|i} A(a) \otimes A'(a) \quad B_j = \sum_b \beta_{b|j} B(b) \otimes B'(b)$$

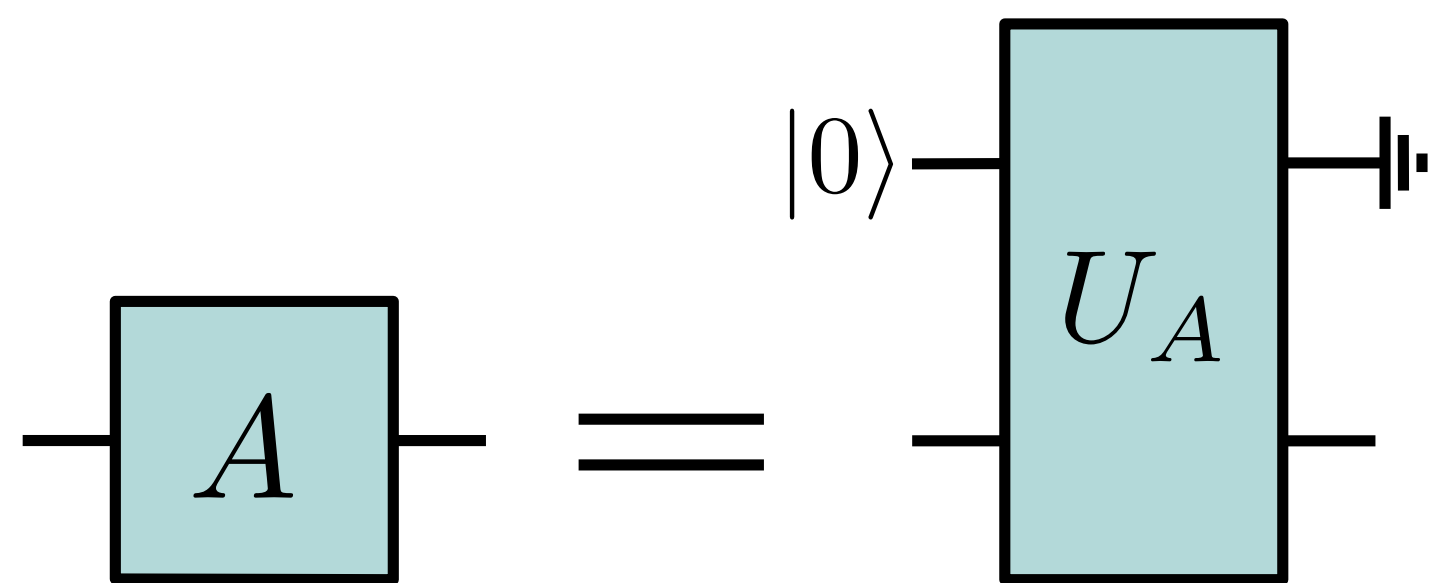
Our simulation



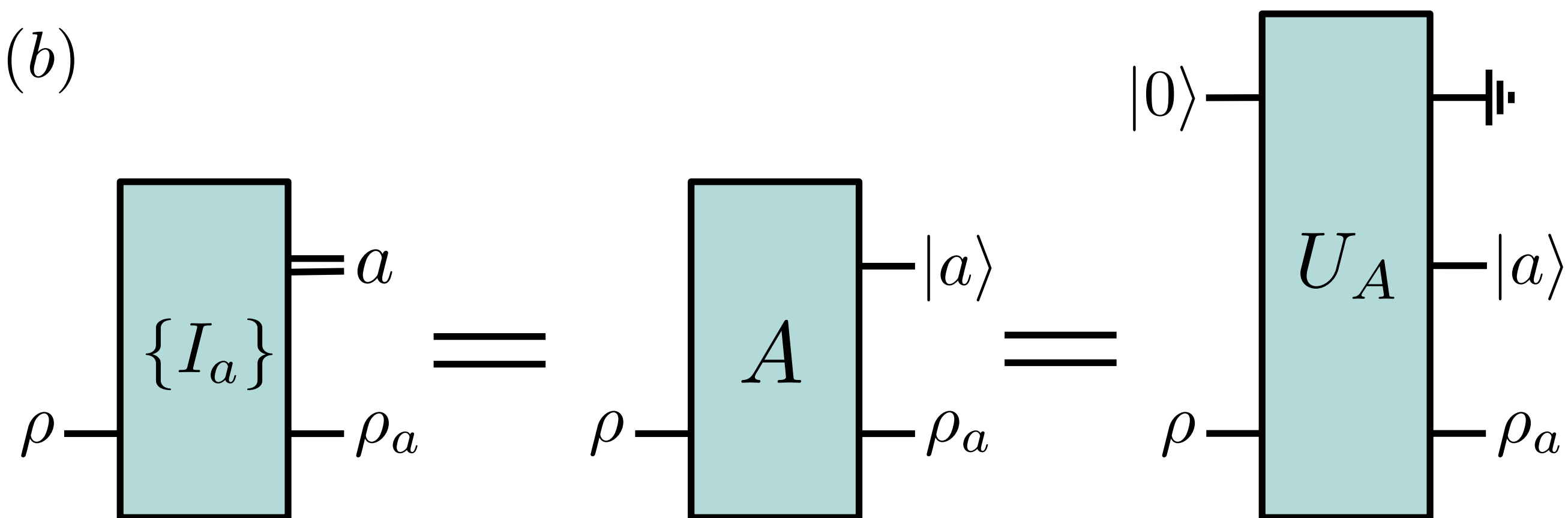
$$C_{ijk} = |0\rangle\langle\otimes|A_k \otimes \sum_{ab} \alpha_{a|i} \beta_{b|j} A'(a) \otimes B(b)A(a) \otimes B'(b) + |1\rangle\langle\otimes|A_i \otimes \sum_{ab} \alpha_{a|k} \beta_{b|j} A'(a) \otimes A(a)B(b) \otimes B'(b)$$

$$A_i = \sum_a \alpha_{a|i} A(a) \otimes A'(a) \quad B_j = \sum_b \beta_{b|j} B(b) \otimes B'(b)$$

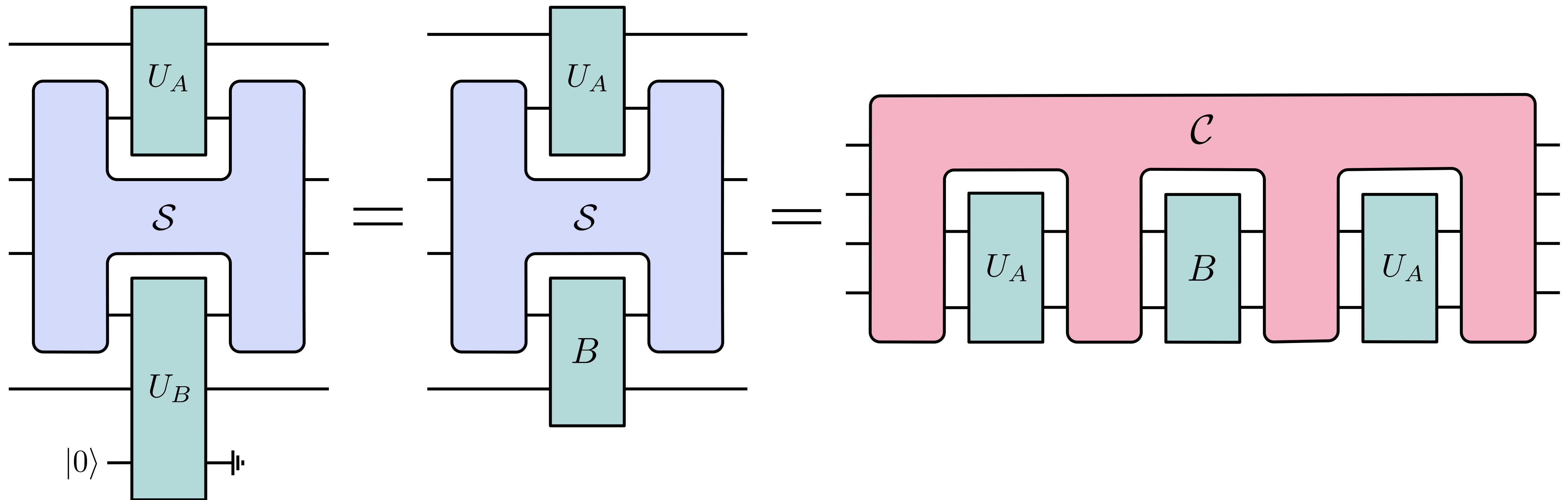
(a)

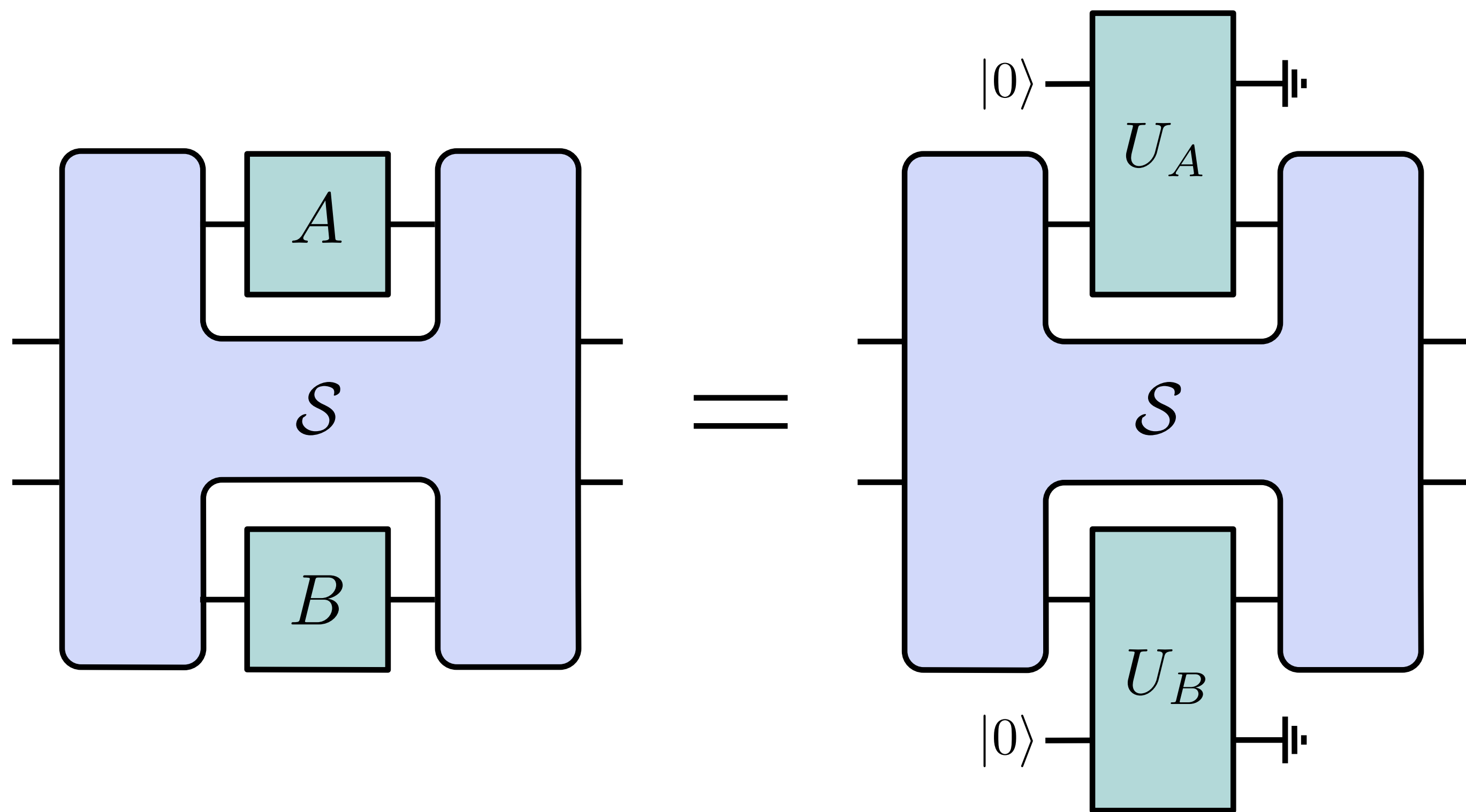


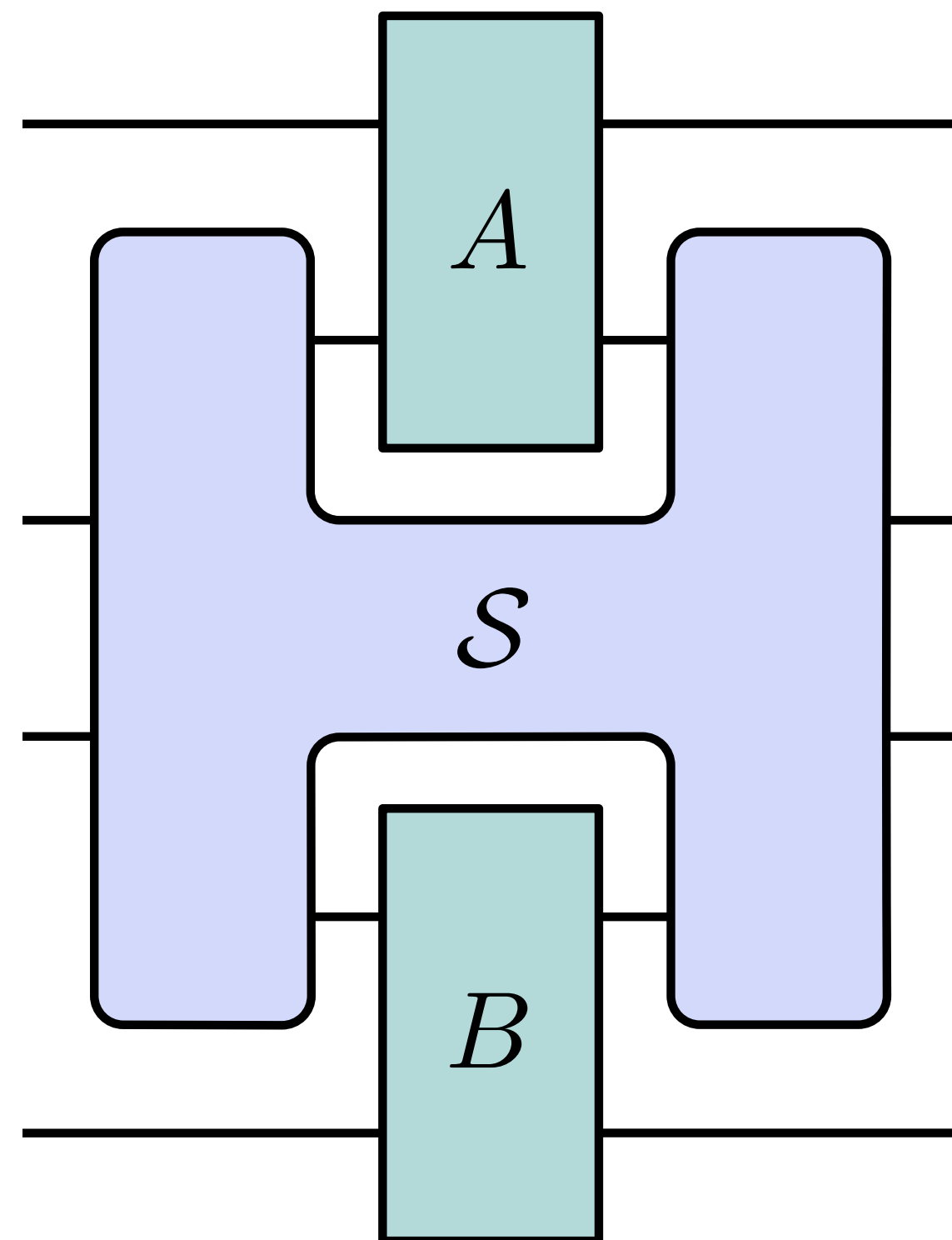
(b)



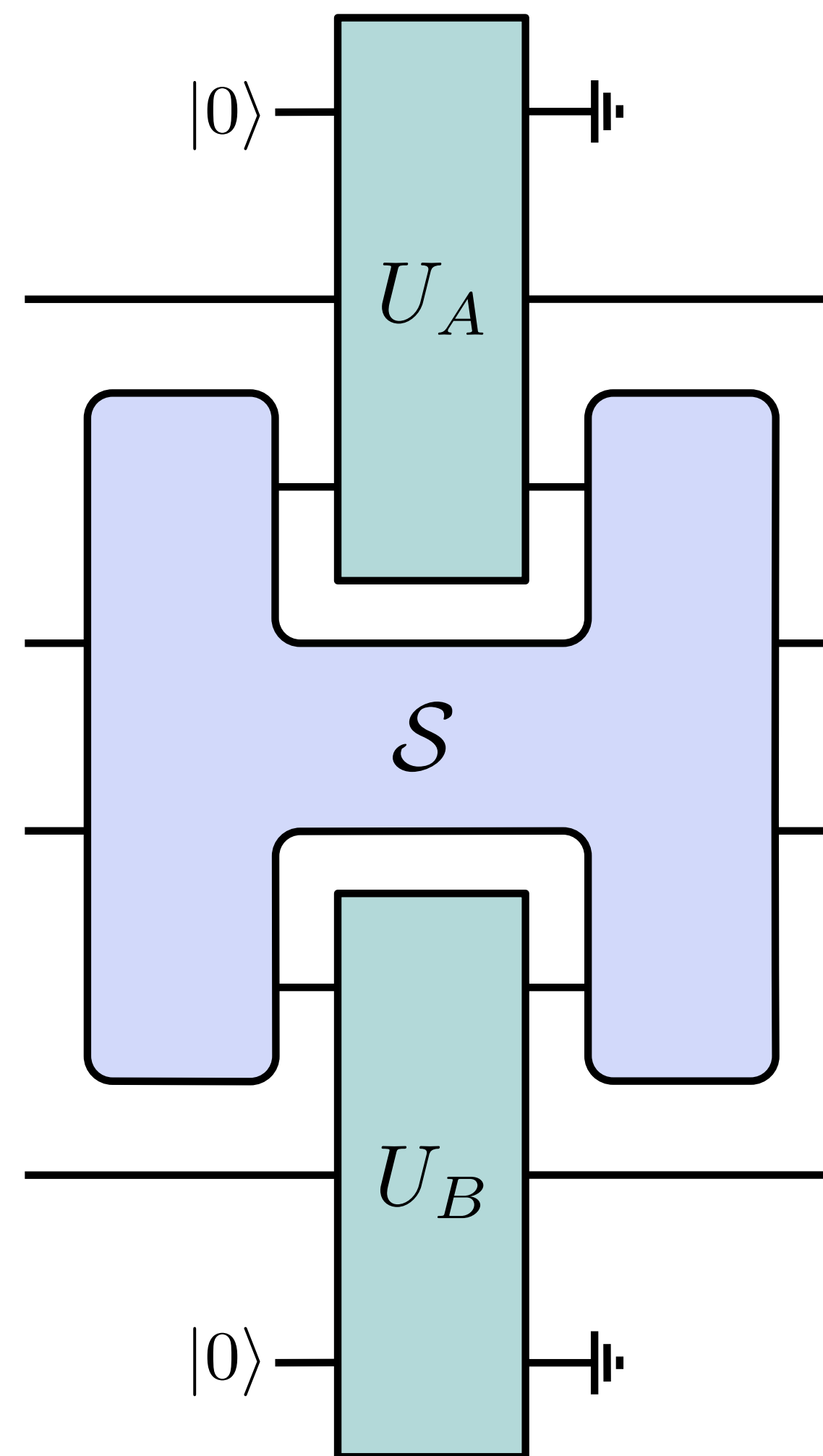
First result: does not generalize of unitary dilations

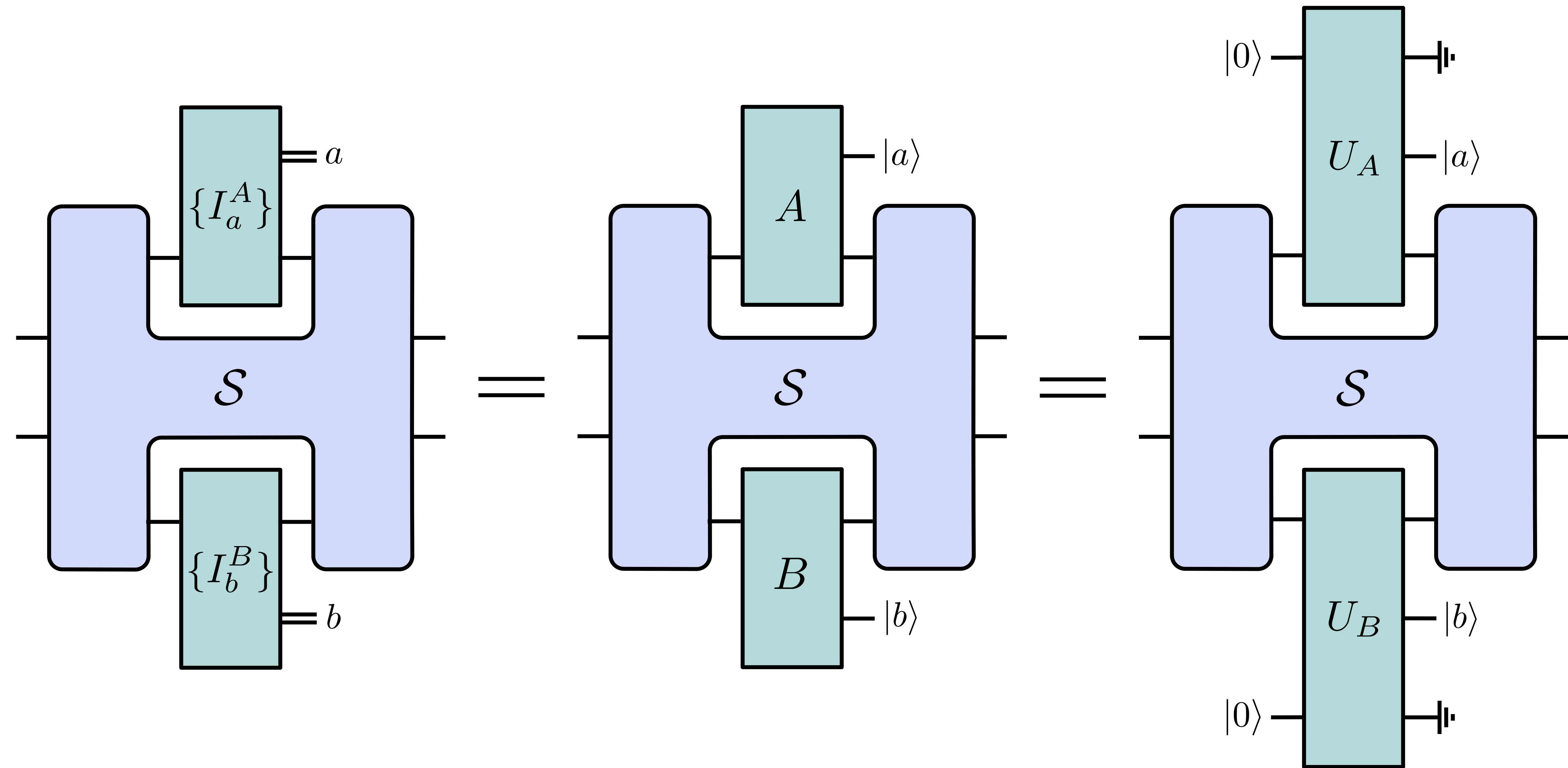


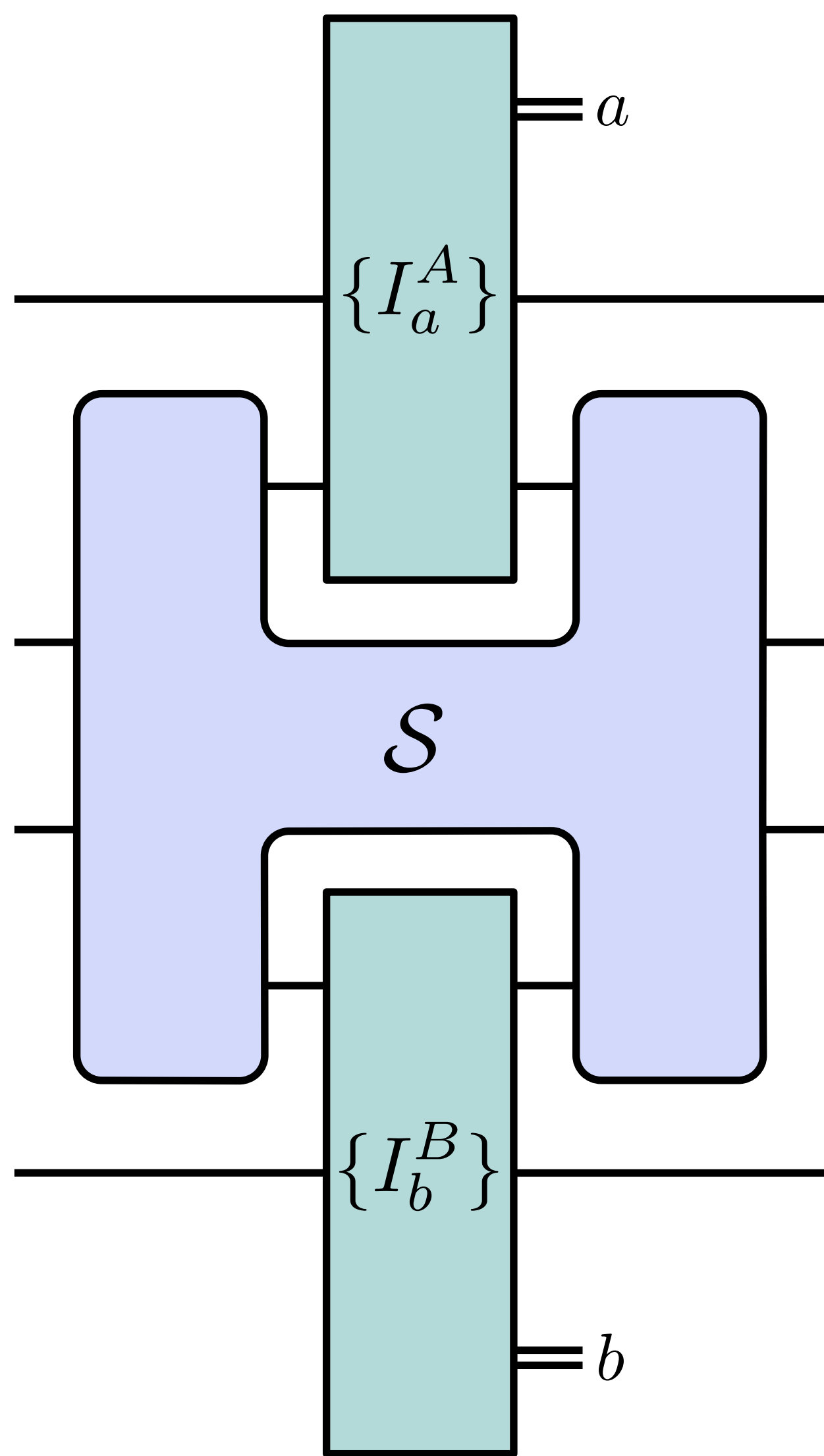




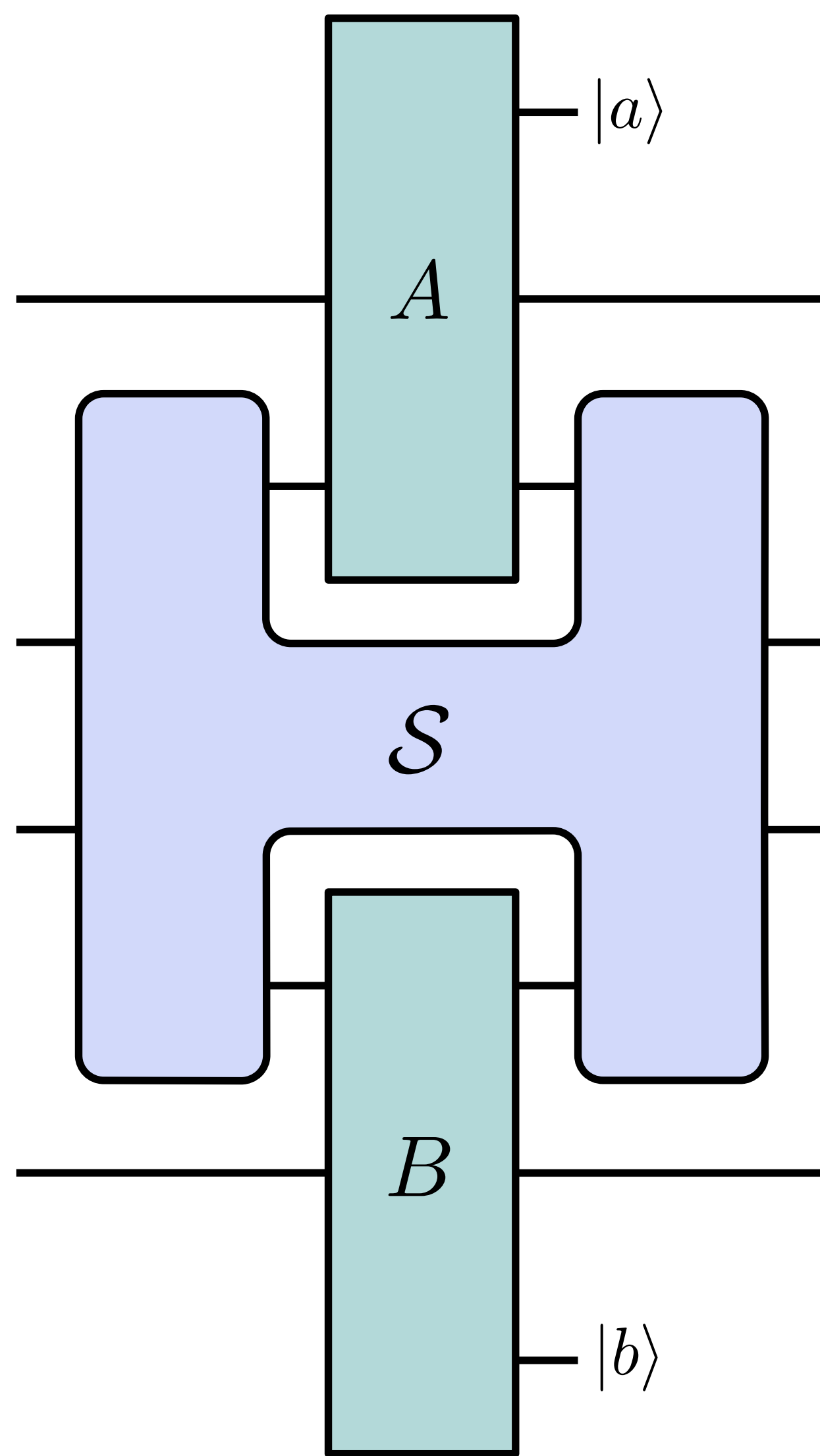
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